



MSc Thesis in Mathematics

Abelianizations of mapping class groups

A calculation of homology of moduli spaces of manifolds

Jan McGarry Furriol

Advisor: Søren Galatius

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ABSTRACT

We study the application [GR16] by Galatius and Randal-Williams of their work on homology of moduli spaces of manifolds to the calculation of abelianizations of the mapping class groups of the manifolds $W_g^{2n} = \#^g(S^n \times S^n)$ for $n \geq 3$ and $g \geq 5$. We use the same approach to carry out these calculations for smooth hypersurfaces of $\mathbb{CP}^4$; the result is that the abelianizations of their mapping class groups are trivial.

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INTRODUCTION

The topological group $\text{Diff}^+(W)$ of orientation preserving diffeomorphisms of a closed oriented smooth manifold W has a classifying space $B\text{Diff}^+(W)$, so that homotopy classes of maps from a space X into $B\text{Diff}^+(W)$ are in natural bijection with fiber bundles with structure group $\text{Diff}^+(W)$ and fiber W . For this reason, we wish to understand the (co)homology of $B\text{Diff}^+(W)$.

The work of Søren Galatius and Oscar Randal-Williams [GR14; GR18; GR17] accomplished an understanding of this homology in terms of homotopy theoretic objects, in the case that W is

simply connected and of dimension $2n \geq 6$, and in a range of degrees depending on the genus

$$g(W) = \max\{g \geq 0 \mid \text{there exists a manifold } M \text{ and a diffeomorphism } W \cong M \sharp (S^n \times S^n)^{\sharp g}\}.$$

In this thesis, we study an application [GR16] of the cited series of papers on homology of moduli spaces of manifolds, which consists in calculating the integral homology $H_1(B\text{Diff}^+(W); \mathbb{Z})$ for certain manifolds W . There is a map to the *mapping class group* of W ,

$$\text{Diff}^+(W) \longrightarrow \pi_0(\text{Diff}^+(W)),$$

which is an isomorphism after abelianizing. Therefore, we can view the computations as that of the abelianization of the mapping class group of W .

Concretely, we first give an exposition of how Galatius and Randal-Williams apply in [GR16] the theory from their series of papers. The manifolds considered there are g -fold connected sums

$$W_g^{2n} = \sharp^g(S^n \times S^n)$$

for natural numbers $n, g \geq 0$. The main result is expressed in terms of the automorphisms $\text{Aut}(Q_{W_g^{2n}})$ of a refinement of the intersection form on the middle homology of W_g^{2n} , and a cobordism group $\Omega_{2n+1}^{(n)}$.

Theorem 0.1 ([GR16]¹). *For all natural numbers $n \geq 3$ and $g \geq 0$ there is a surjective map*

$$H_1(B\Gamma_g^n; \mathbb{Z}) \longrightarrow \Omega_{2n+1}^{(n)} \oplus H_1(B\text{Aut}(Q_{W_g^{2n}}); \mathbb{Z}).$$

from the abelianization of the mapping class group $\Gamma_g^n = \pi_0(\text{Diff}^+(W_g^{2n}))$. For $g \geq 5$ it is an isomorphism, and, furthermore,

$$H_1(B\text{Aut}(Q_{W_g^{2n}}); \mathbb{Z}) \cong \begin{cases} (\mathbb{Z}/2)^2 & n \text{ even,} \\ 0 & n = 1, 3, \text{ or } 7, \\ \mathbb{Z}/4 & n \text{ otherwise.} \end{cases}$$

The cobordism group $\Omega_{2n+1}^{(n)}$ is a quotient of the cokernel of the J -homomorphism, a map to the stable homotopy groups of spheres $J: \pi_*(O) \rightarrow \pi_*^s$. As long as $n > 1$ and $n \neq 8, 11$, it is isomorphic to the cokernel of J , by work of various authors. (See section 2.2.)

Our focus in the exposition will be to examine the map to the cobordism group $\Omega_{2n+1}^{(n)}$, as it is where the mentioned theory comes into play.

Secondly, we consider smooth hypersurfaces $V \subset \mathbb{CP}^4$ and use the same approach in order to calculate the abelianizations of their mapping class groups. This is a family of examples considered in [GR20, §5.3], where the rational cohomology $H^*(B\text{Diff}^+(V); \mathbb{Q})$ is calculated. We incorporate our own calculations (section 3.5), and one by [BG87], and we obtain the following result in terms of the genus $g(V)$, which is computed in section 3.3 in terms of the degree of V . A group is called *perfect* if its abelianization is trivial.

Proposition 0.2. *Let $d \geq 1$ be an integer; and $V_d \subset \mathbb{CP}^4$ be a smooth hypersurface of degree d . Let $g \geq \max\{5 - g(V_d), 0\}$. Then the mapping class group $\pi_0(\text{Diff}^+(V_d \sharp (S^3 \times S^3)^{\sharp g}))$ is perfect.*

¹Theorem 0.1 is a part of the main theorem in this reference.

Organization. Section 1 is dedicated to the presentation of the theory from [GR14; GR18; GR17], as well as that of its origins and prerequisites. In section 2, we give an exposition of the application of this theory in [GR16]; the main result is Theorem 0.1. Finally, in section 3 we take the same approach as in [GR16] in order to prove Proposition 0.2.

Conventions By *space* we mean a compactly generated weakly Hausdorff topological space. By *smooth manifold* we mean a Hausdorff, second countable topological manifold equipped with a C^∞ maximal atlas. By a *smooth map* we mean a C^∞ map. All manifolds shall be smooth. Unless stated otherwise, vector bundles are over the real numbers.

1. PARAMETRIZED PONTRJAGIN–THOM THEORY

The goal of this section is to state and explain the main theorem 1.10, which we will apply in sections 2 and 3, while introducing concepts that will appear throughout this thesis.

The main theorem establishes a relation between geometry and homotopy theory which stems from the Pontrjagin–Thom correspondence 1.2. The recipients of this correspondence on the side of homotopy theory are Thom spaces and Thom spectra, introduced in section 1.1. On the side of geometry there is bordism theory, the subject of section 1.3.

The instance of this in the main theorem is a parametrized version of it, that is, for bundles over a space, and it is covered in sections 1.4 and 1.5. Finally, we will reach the main theorem.

1.1. Thom spectra and orientations

The following is the key object in the Pontrjagin–Thom correspondence.

Definition 1.1. Let $\xi: V \rightarrow X$ be a rank n vector bundle, and $S^\xi \rightarrow X$ the fiber bundle obtained by applying one-point compactification to each fiber of ξ . The *Thom space* of ξ is the quotient space $\text{Th}(\xi) = S^\xi/X$ of the section $X \rightarrow S^\xi$ given by the points at infinity.

We have referred to the term *classifying space* of a group G , which is a space BG together with a principal G -bundle $EG \rightarrow BG$ such that, for any space X , the pullback operation establishes a natural bijection between homotopy classes of maps $[X, BG]$ and isomorphism classes of principal G -bundles over X . Such a classifying space is unique up to homotopy equivalence, and one construction of it is outlined in Appendix 4.1. In this case of the orthogonal group $O(n)$, we fix the model $BO(n) = \text{Gr}_n(\mathbb{R}^\infty) = \text{colim}_r \text{Gr}_n(\mathbb{R}^{n+r})$ with $EO(n)$ the frame bundle, where $\text{Gr}_n(\mathbb{R}^{n+r})$ denotes the Grassmannian of n -planes in $\mathbb{R}^{n+r}$, and the maps in the colimit are induced by the canonical maps $\mathbb{R}^r \rightarrow \mathbb{R}^r \times \mathbb{R} = \mathbb{R}^{r+1}$. We let $BO = \text{colim}_n BO(n)$.

Now, let us assume that X is a path connected space. Otherwise, one considers one path component at a time.

Definition 1.2. A *stable vector bundle* on X is a map $X \rightarrow \mathbb{Z} \times BO$. Its rank is the integer in the image of the locally constant map $X \rightarrow \mathbb{Z} \times BO \rightarrow \mathbb{Z}$.

Whitney sum gives a map $(\mathbb{Z} \times BO) \times (\mathbb{Z} \times BO) \rightarrow \mathbb{Z} \times BO$, and the homeomorphism of Grassmannians $\iota_{d,n}: \text{Gr}_d(\mathbb{R}^{d+n}) \rightarrow \text{Gr}_n(\mathbb{R}^{d+n})$ sending a d -plane to its orthogonal complement in $\mathbb{R}^{n+d}$ gives an inversion map $\iota: \mathbb{Z} \times BO \rightarrow \mathbb{Z} \times BO$ on the colimit. For a general stable vector bundle, we will be able to define a Thom spectrum.

Spectra. A *spectrum* is a sequence of pointed spaces X_n , indexed by the integers, together with basepoint-preserving maps $\sigma_n: \Sigma X_n \rightarrow X_{n+1}$. A map $f: X \rightarrow Y$ of spectra consists of a sequence of basepoint-preserving maps $f_n: X_n \rightarrow Y_n$, which are compatible with the structure maps σ_n in the sense that $f_{n+1} \circ \sigma_n = \sigma_n \circ \Sigma f_n$.

Every pointed space X gives rise to the suspension spectrum $\Sigma^\infty X$, whose n th space is $\Sigma^n X$. We will write $S = \Sigma^\infty S^0$ for the *sphere spectrum*, and $S^n = \Sigma^\infty S^n$ for its suspensions; it will be clear from the context whether S^n is the n -dimensional sphere or its suspension spectrum. Given a spectrum X , we can define its *infinite loop space* as the pointed space $\Omega^\infty X = \text{colim}_n \Omega^n X_n$, where

the maps in the colimit are the adjoints of the structure maps of the spectrum. This assignment assembles into a functor right adjoint to Σ^∞ .

We define the i th homotopy group of X as the colimit of homotopy groups

$$\pi_i(X) = \operatorname{colim}_n \pi_{i+n}(X_n),$$

where $\pi_{i+n}(X_n) \rightarrow \pi_{i+n+1}(X_{n+1})$ is the composition of the suspension homomorphism and $\pi_{i+n}(\sigma_n)$. Similarly, we define the i th homology group with coefficients in an abelian group A as

$$H_i(X) = \operatorname{colim}_n H_{i+n}(X_n; A).$$

We will sometimes write $\pi_i^s(X) = \pi_i(\Sigma^\infty X)$ if X is a pointed space; in this case, notice that $H_i(\Sigma^\infty X; A) \cong H_i(X; A)$ because of the suspension isomorphism.

Another family of examples is Eilenberg–Mac Lane spectra. For an abelian group A and a natural number n , an *Eilenberg–Mac Lane space* $K(A, n)$ is a space such that

$$\pi_i(K(A, n)) \cong \begin{cases} A & i = n \\ 0 & i \neq n. \end{cases}$$

See Appendix 4.1 for a proof that such spaces exist, and that there are maps $K(A, n) \rightarrow \Omega K(A, n+1)$ – which are in fact homotopy equivalences. These spaces form a spectrum HR whose n th space is $K(A, n)$ and with the adjoint maps $\Sigma K(A, n) \rightarrow K(A, n+1)$.

If R is a ring, then its unit $1 \rightarrow R$ and product $R \otimes R \rightarrow R$ induce maps $\eta: S \rightarrow HR$ and $\mu: HR \wedge HR \rightarrow HR$, giving HR the structure of an algebra object in the stable homotopy category equipped with the smash product $\wedge$, which gives it a symmetric monoidal structure [Ada74]; that is, η and μ satisfy unitality and associativity up to *homotopy*. We refer to loc. cit. for more details.

Thom spectra and orientations. Observe that Thom spaces are canonically pointed, and that through this construction a vector bundle map induces a map of based spaces in a functorial way. We proceed to define Thom spectra for stable vector bundles.

Let $V: X \rightarrow \mathbb{Z} \times BO$ be a stable vector bundle of rank d and consider the projection $v: X \rightarrow BO$. For each natural number n , let X_n be the homotopy pullback

$$\begin{array}{ccc} X_n & \longrightarrow & X \\ \downarrow v_n \lrcorner & & \downarrow v \\ BO(n) & \longrightarrow & BO. \end{array}$$

Let $v_n^* \gamma_n$ be the pullback along v_n of γ_n , the tautological vector bundle of rank n over $BO(n)$. The pullback square

$$\begin{array}{ccc} X_n & \longrightarrow & X_{n+1} \\ \downarrow v_n \lrcorner & & \downarrow v_{n+1} \\ BO(n) & \longrightarrow & BO(n+1) \end{array}$$

induces a map of vector bundles $^2 v_n^* \gamma_n \oplus \underline{\mathbb{R}} \rightarrow v_{n+1}^* \gamma_{n+1}$, induced via pullback by the bundle map $\gamma_n \oplus \underline{\mathbb{R}} \rightarrow \gamma_{n+1}$ covering $BO(n) \rightarrow BO(n+1)$. For any vector bundle ξ , there is a basepoint-preserving homeomorphism $\Sigma \operatorname{Th}(\xi) \cong \operatorname{Th}(\xi \oplus \underline{\mathbb{R}})$. Therefore, the sequence of spaces $\operatorname{Th}(v_n^* \gamma_n)$

² $\underline{\mathbb{R}}$ denotes the trivial vector bundle of rank 1.

together with the maps $\sigma_n: \Sigma \text{Th}(v_n^* \gamma_n) \rightarrow \text{Th}(v_{n+1}^* \gamma_{n+1})$ form a spectrum. The following is a shift of it according to the dimension d of the stable vector bundle.

Definition 1.3. Let $V: X \rightarrow \mathbb{Z} \times BO$ be a stable vector bundle of rank d . Its *Thom spectrum* $\text{Th}(V)$ is the spectrum whose $(n-d)$ th space is $\text{Th}(v_n^* \gamma_n)$, with structure maps σ_n .

For example, if a vector bundle $\xi: V \rightarrow X$ of rank $d \geq 0$ is classified by a map $X \rightarrow BO(d)$, then the Thom spectrum of $X \rightarrow \{d\} \times BO(d) \rightarrow \mathbb{Z} \times BO$ is the suspension spectrum $\Sigma^\infty \text{Th}(\xi)$.

Let R be a ring, and $V: X \rightarrow \mathbb{Z} \times BO$ a stable vector bundle of rank d . A *Thom class* or *R-orientation* of V is a cohomology class $u \in H^d(\text{Th}(V); R)$ ³ such that for any $x \in X$, the image of u under the map induced by the restriction of the fiber $V_x: \{x\} \rightarrow X \rightarrow \mathbb{Z} \times BO$

$$H^d(\text{Th}(V); R) \longrightarrow H^d(\text{Th}(V_x); R) \cong H^d(S^d) \cong R$$

is a unit. We obtain an isomorphism $H^*(X; R) \rightarrow H^{*+d}(\text{Th}(V); R)$ by sending $x \in H^n(X; R)$ to

$$x \cdot u: \text{Th}(V) \longrightarrow \Sigma^\infty X_+ \wedge \text{Th}(V) \xrightarrow{x \wedge u} \Sigma^n HR \wedge \Sigma^d HR \longrightarrow \Sigma^{n+d} HR. \quad (1)$$

See Appendix 4.3 for a proof.

1.2. Tangential structures

Let $V \rightarrow X$ and $U \rightarrow Y$ be two vector bundles. A *bundle map* is a continuous map $V \rightarrow U$ of total spaces of the vector bundles, which on each fiber of V restricts to a linear isomorphism onto a fiber of U . Let $\text{Bun}(V, U) \subset \text{Map}(U, V)$ denote the space of all bundle maps equipped with the compact-open topology.

Definition 1.4. Let W be a d -dimensional manifold and $\theta: B \rightarrow BO(d)$ be a map. A (*tangential*) θ -*structure* on W is a bundle map $\hat{l}: TW \rightarrow \theta^* \gamma_d$.

Group homomorphisms $G \rightarrow \text{GL}_d(\mathbb{R})$ give examples by considering the map on classifying spaces $\theta: BG \rightarrow B\text{GL}_d(\mathbb{R}) \simeq BO(d)$. If we take $G = \text{SO}(d) \subset \text{O}(d)$, then a θ -structure is an orientation. We will also consider $G = \text{Spin}(d)$ the connected double cover of $\text{SO}(d)$.

These examples fit in homotopy fiber sequences⁴

$$BSO(d) \xrightarrow{\theta^+} BO(d) \xrightarrow{w_1} K(\mathbb{Z}/2, 1),$$

where w_i is a map representing the i th Stiefel–Whitney class⁵; and

$$B\text{Spin}(d) \longrightarrow BSO(d) \xrightarrow{w_2} K(\mathbb{Z}/2, 2).$$

So, for example, the obstruction to lift a map $u: X \rightarrow BO(d)$ to a map $X \rightarrow BSO(d)$ along θ^+ (that is, to provide $u^* \gamma_d \rightarrow X$ with an orientation) is precisely the first Stiefel–Whitney class⁶.

³We define the cohomology of a Thom spectrum as $H^i(\text{Th}(V); R) = \lim_n H^{i+n}(\text{Th}(V)_n; R)$. This is not well-defined for an arbitrary spectrum, but it is for Thom spectra (see Remark 4.5).

⁴See Appendix 4.1 for the definition.

⁵See Appendix 4.4 on characteristic classes.

⁶Obstruction theory is studied in section 3.6.1.

Once chosen such a lift, if it exists, the obstruction for a spin structure is precisely the second Stiefel–Whitney classes. We will also consider the homotopy fiber

$$B\text{Spin}^c(d) \longrightarrow B\text{SO}(d) \times K(\mathbb{Z}, 2) \xrightarrow{w_2 \times \text{red}_2} K(\mathbb{Z}/2, 2)$$

of the product of the second Stiefel–Whitney class and the map induced by the reduction modulo 2. We record Lemma 1.1 for later reference.

Lemma 1.1. *Let $u_X: X \rightarrow \text{BO}(d)$ and $u_Y: Y \rightarrow \text{BO}(d)$ be maps of spaces, and suppose that X is a CW complex. Let $U_X = u_X^* \gamma_d$ and $U_Y = u_Y^* \gamma_d$ be the pullbacks of the tautological bundle γ_d over $\text{BO}(d)$. The canonical map $\text{Map}_{/\text{BO}(d)}(u_X, u_Y) \rightarrow \text{Bun}(U_X, U_Y)$ from the space of maps over $\text{BO}(d)$ is a weak equivalence.*

Proof. For the universal case $u_Y = \text{id}_{\text{BO}(d)}$, it is the inclusion of a point in a contractible space; see, for example, [Gal+09, Lemma 5.1]. In general, we rewrite the map as

$$\{u_X\} \times_{\text{Map}(X, \text{BO}(d))} \text{Map}(X, Y) \longrightarrow \text{Bun}(U_X, \gamma_d) \times_{\text{Map}(U_X, \text{BO}(d))} \text{Map}(U_X, Y).$$

The maps between pullback diagrams are all weak equivalences: $\{u_X\} \rightarrow \text{Bun}(U_X, \gamma_d)$ is the universal case, and the weak equivalence $U_X \rightarrow X$ induces one on mapping spaces assuming X is cofibrant (by [GS07, 4.11] and Ken Brown’s Lemma). $\square$

1.3. Bordism theory

We refer to [Sto68] and [MW65] for more details of the discussion that follows.

Let $\theta: B \rightarrow \text{BO}$ be a fibration and $d \geq 0$. The bordism group Ω_d^θ consists of bordism classes of triples (M, l, ν_M)

$$\begin{array}{ccc} & & B \\ & \nearrow l & \downarrow \theta \\ M & \xrightarrow{\nu_M} & \text{BO} \end{array}$$

of a smooth closed d -manifold M , the stable normal Gauss map ν_M of M for some embedding $i: M \rightarrow \mathbb{R}^{d+n}$, and a lift l making the triangle commute. If n is sufficiently large, then the Gauss maps of any two such embeddings are homotopic, and therefore yield the same bordism class. For this reason, we will omit it in the notation.

Such an l is a *stable normal θ -structure*. The homeomorphism $\iota_{d,n}: \text{Gr}_d(\mathbb{R}^{d+n}) \rightarrow \text{Gr}_n(\mathbb{R}^{d+n})$ induces a map $\iota: \text{BO} \rightarrow \text{BO}$ such that $\iota^2 = \text{id}$. Given an embedding $M^d \rightarrow \mathbb{R}^{n+d}$, the Gauss maps $\nu_M: M \rightarrow \text{Gr}_n(\mathbb{R}^{n+d}) \hookrightarrow \text{BO}$ and $\tau_M: M \rightarrow \text{Gr}_d(\mathbb{R}^{n+d}) \hookrightarrow \text{BO}$ correspond to each other under $\iota_{d,n}$, that is $\nu_M = \iota_{d,n} \tau_M$. In this way, stable normal $-\theta = \iota^* \theta$ -structures are in natural bijection with *stable tangential θ -structures*, that is lifts $M \rightarrow B$ along θ of the stable tangent bundle τ_M of the embedding.

1.3.1 The Pontrjagin–Thom theorem

We view $S^n = \mathbb{R}^n \cup \{\infty\}$ as the one-point compactification of $\mathbb{R}^n$, with ∞ as a basepoint.

Definition 1.5. Let M_0, M_1 be compact codimension d submanifolds of $S^n - \infty$, and without boundary. Then a *bordism from M_0 to M_1 within S^n* is a compact manifold $W \subset (S^n - \infty) \times [0, 1]$ of codimension d such that for some $\epsilon > 0$ we have:

- $W \cap (S^n \times [0, \epsilon)) = M_0 \times [0, \epsilon),$
- $W \cap (S^n \times (\epsilon, 1]) = M_1 \times (\epsilon, 1],$ and
- $\partial W = W \cap (S^n \times \{0, 1\}).$

We say that M_0 is *cobordant to M_1 within S^n* .

Definition 1.6. Let $\Theta: V \rightarrow X$ be a vector bundle. Suppose that the two submanifolds are equipped with Θ -structures $\phi_i: N_{M_i}^{S^n} \rightarrow V$ on their normal bundles; consider the pairs (M_i, ϕ_i) , $i = 0, 1$. We say that (M_0, ϕ_0) and (M_1, ϕ_1) are *Θ -cobordant within S^n* if there exists a bordism W from M_0 to M_1 within S^n , and a Θ -structure $\varphi: N_W^{S^n \times [0, 1]} \rightarrow V$ such that the restriction of φ to M_i is the bundle map $\varphi|_{M_i} = \phi_i: N_{M_i}^{S^n \times [0, 1]}|_{M_i} = N_{M_i}^{S^n} \rightarrow V$ for $i = 0, 1$.

Consider the set

$$\Omega_{n-d; S^n}^\Theta = \{(M, \phi) \mid M^{n-d} \subset S^n - \infty \text{ compact without boundary, } \phi: N_M^{S^n} \rightarrow V\} / \sim$$

where $\sim$ is the equivalence relation of being Θ -cobordant within S^n .

Theorem 1.2 (Pontrjagin–Thom). *Let $V \rightarrow X$ be a smooth vector bundle of rank $d \geq 0$. For all $n \geq d$, there is a bijection*

$$\pi_n(\text{Th}(V), \infty) \cong \Omega_{n-d; S^n}^\Theta$$

which is natural with respect to maps of vector bundles.

Proof sketch. Let $g: S^n \rightarrow \text{Th}(V)$ be a map. Observe that the zero section embeds $X \subset V$ as a smooth submanifold of codimension d . By [MS16, 18.6], g can be homotoped to a map $f: S^n \rightarrow \text{Th}(V)$ which is smooth throughout $f^{-1}(\text{Th}(V) - \infty)$ and transverse to the zero section X . Then, by the inverse function theorem, $f^{-1}(X) \subset S^n$ is a submanifold of dimension $n - d$ (see [MW65, p. 11]). Moreover, at each point $x \in f^{-1}(X)$, the linear subspace $T_x f^{-1}(X) \subset T_x S^n$ maps to zero under the quotient $df_x: T_x S^n \rightarrow T_{f(x)} V \rightarrow T_{f(x)} V / T_{f(x)} X$. Thus there is a canonical linear map between the normal vector spaces $df_x = \phi_x: T_x S^n / T_x f^{-1}(X) \rightarrow T_{f(x)} V / T_{f(x)} X$, which is surjective by transversality; since these are of the same finite dimension (d), we deduce that it is a linear isomorphism. These assemble into a vector bundle map $\phi: N_{f^{-1}(X)}^{S^n} \rightarrow N_X^V$ over $f|_{f^{-1}(X)}$. Note that $N_X^V \cong V$ as vector bundles; also denote by ϕ the composite to V . Consider the pair $(f^{-1}(X), \phi)$.

The above construction is homotopy invariant, because if $H: S^n \times [0, 1] \rightarrow \text{Th}(V)$ is a homotopy between two *smooth* maps as above, then H can be homotoped to a map which is smooth throughout the inverse image of $\text{Th}(V) - \infty$ and transverse to X . Importantly, this homotopy of H can be assumed constant near the endpoints $S^n \times \{0, 1\}$, so that the resulting smooth homotopy is one between the original maps. Applying the above construction to the smooth homotopy yields a cobordism between the corresponding pairs, so we have a well-defined map

$$\pi_n(\text{Th}(V)) \longrightarrow \Omega_{n-d; S^n}^\Theta, \quad [f] \longmapsto [f^{-1}(X), \phi].$$

For surjectivity, consider a pair (M, ϕ) and choose a tubular neighborhood $U \cong N_M^{S^n}$ of $M \subset S^n - \infty$. The composite $U \rightarrow V$ is transverse to the zero section X , whose preimage under this map is M . We can extend the map $U \rightarrow V$ to a basepoint-preserving map

$$S^n \longrightarrow \text{Th}(V) \tag{2}$$

by sending $S^n - U$ to ∞ . We have given a preimage of (M, ϕ) . For injectivity, this same construction applied to a cobordism yields a homotopy. $\square$

Remark 1.3. The construction 2 is called a *Pontrjagin–Thom collapse map*.

Let $\theta : B \rightarrow BO$ be a fibration. Consider the pullback $\theta_d : B(d) \rightarrow BO(d)$ and $V = \theta^* \gamma_d$. Let $[M^n, \phi] \in \Omega_{n; S^{n+d}}^{\theta_d}$. Consider the bundle maps

$$\begin{array}{ccccc} N_M^{S^{n+d}} & \xrightarrow{\phi} & \theta_d^* \gamma_d & \longrightarrow & \gamma_d \\ \downarrow & & \downarrow & & \downarrow \\ M & \xrightarrow{\bar{\phi}} & B(d) & \longrightarrow & BO(d). \end{array}$$

If d is large (depending only on n), then the map of base spaces $M \rightarrow BO(d)$ is homotopic to the Gauss map $\nu_M : M \rightarrow BO(d)$ of $M \subset S^{n+d}$, as they both classify the normal bundle. Since θ_d is a fibration, we may cover this homotopy by a homotopy of $\bar{\phi}$ to a new map l , so that the latter is a lift along θ_d of the Gauss map. Consider the stabilization $l : M \rightarrow BO(d) \rightarrow BO$ and the pair $[M, l] \in \Omega_n^\theta$. We get an isomorphism

$$\Omega_n^\theta \xleftarrow{\cong} \operatorname{colim}_d \Omega_{n; S^{n+d}}^{\theta_d}$$

where the maps in the colimit are induced by the inclusions $\mathbb{R}^r \rightarrow \mathbb{R}^r \times \mathbb{R}$ after one-point compactifying. (For well-definedness, follow the same construction with a bordism in the place of M . By considering the bundle map $l^* \theta^* \gamma_d \rightarrow \theta_d^* \gamma_d$ induced on pullbacks, surjectivity is clear. For injectivity, use the surjectivity argument on a bordism and observe that pulling back $\theta_d^* \gamma_d$ along $\bar{\phi}$ we recover ϕ . So ϕ is bundle homotopic to $l^* \theta^* \gamma_d \rightarrow \theta_d^* \gamma_d$.) Inspecting the proof of Theorem 1.2, we see that it gives a bijection $\Omega_{n; S^{n+d}}^{\theta_d} \cong \pi_{n+d}(\operatorname{Th}(\theta_d^* \gamma_d))$ using the smoothness of γ_d . Observe that if d is large, then two n -dimensional submanifold of S^{n+d} may be isotoped to be disjoint. Disjoint union gives $\Omega_{n; S^{n+d}}^{\theta_d}$ a group operation, and the map in 1.2 is a group homomorphism.

Corollary 1.4. *Let $\theta : B \rightarrow BO$ be a fibration. For all $d \geq 0$ there is a group isomorphism*

$$\Omega_n^\theta \cong \operatorname{colim}_d \pi_{n+d}(\operatorname{Th}(\theta_d^* \gamma_d))$$

which is natural with respect to maps of stable vector bundles.

Remark 1.5. If $\theta : B \rightarrow BO$ is a stable vector bundle of rank $r \in \mathbb{Z}$, then this gives

$$\Omega_n^\theta \cong \pi_{n+r}(\operatorname{Th}(\theta)).$$

1.4. The parametrized Pontrjagin–Thom map

Let W be a closed d -manifold. The group $\operatorname{Diff}(W)$ of diffeomorphisms becomes a topological group when topologized with the C^∞ -topology ([Hir12]). We consider fiber bundles $\pi : E \rightarrow X$ with structure group $\operatorname{Diff}(W)$ and fiber W . We refer to such bundles as W -bundles. Let $\pi : E \rightarrow X$ be a W -bundle and $P \rightarrow X$ the corresponding principal $\operatorname{Diff}(W)$ -bundle. The *vertical tangent bundle* of π is the vector bundle

$$T^\vee E = P \times_{\operatorname{Diff}(W)} TW \longrightarrow P \times_{\operatorname{Diff}(W)} W \cong E.$$

For the construction of the parametrized Pontrjagin–Thom map, we will need a tubular neighborhood of each fiber E_x for some embedding of it into Euclidean space. Suppose that X is paracompact.

Consider the space of embeddings $\text{Emb}(W, \mathbb{R}^n)$ equipped with the C^∞ -topology. The inclusions $\mathbb{R}^n \subset \mathbb{R}^n \times \mathbb{R}$ induce maps between these spaces, and we denote their colimit by $\text{Emb}(W, \mathbb{R}^\infty)$. Since the space $\text{Emb}(W, \mathbb{R}^\infty)$ is weakly contractible by Whitney’s embedding theorem, a section of $P \times_{\text{Diff}(W)} \text{Emb}(W, \mathbb{R}^\infty) \rightarrow X$ always exists, by obstruction theory. We will call such a section a *fiberwise embedding*, and it is equivalently given by a $\text{Diff}(W)$ -equivariant continuous map $P \rightarrow \text{Emb}(W, \mathbb{R}^\infty)$; the restriction i_x of its adjoint $i: E \rightarrow \mathbb{R}^\infty$ to a fiber E_x is an embedding. We say that an embedding $W \rightarrow \mathbb{R}^n$ is *fat* if the exponential map $N \rightarrow \mathbb{R}^n$ from the normal bundle of the embedding restricts to a diffeomorphism on the unit disc bundle $D(N)$. We will consider *fiberwise fat embeddings*, that is sections of $P \times_{\text{Diff}(W)} \text{Emb}^{\text{fat}}(W, \mathbb{R}^n) \rightarrow X$ where $\text{Emb}^{\text{fat}}(W, \mathbb{R}^n) \subset \text{Emb}(W, \mathbb{R}^n)$ is the weakly equivalent subspace consisting of fat embeddings.

Suppose that we have $i: E \rightarrow X \times \mathbb{R}^n$ a fiberwise fat embedding, and consider the fiberwise normal bundle $N_i^\vee \rightarrow E$. By the fatness assumption, the exponential map $D(N_i^\vee) \rightarrow U$ is fiberwise a diffeomorphism onto a fiberwise tubular neighborhood U of E in $X \times \mathbb{R}^n$. By collapsing the complement of U , we get a map

$$X_+ \wedge S^n \longrightarrow \text{Th}(N_i^\vee).$$

Now let $i: E \rightarrow X \times \mathbb{R}^\infty$ be a fiberwise fat embedding; it provides a Gauss map $\tau^\vee E: E \rightarrow \text{BO}(d)$ for $T^\vee E$. By applying the above construction on the subspaces $X_n = \{x \in X \mid i_x(E_x) \subset \{x\} \times \mathbb{R}^n\}$ and taking the colimit with respect to the inclusions we get a map of spectra

$$\Sigma^\infty X_+ \longrightarrow \mathbf{Th}(-\tau^\vee E).$$

Any two sections of $P \times_{\text{Diff}(W)} \text{Emb}^{\text{fat}}(W, \mathbb{R}^\infty)$ are homotopic because $\text{Emb}^{\text{fat}}(W, \mathbb{R}^\infty)$ is weakly contractible. Therefore, another choice of fiberwise fat embedding yields a map $\Sigma^\infty X_+ \rightarrow \mathbf{Th}(-\tau^\vee E)$ in the same homotopy class.

Now let $\theta: B \rightarrow \text{BO}(d)$ be a fibration, and suppose that we have a θ -structure $\hat{l}: T^\vee E \rightarrow \theta^* \gamma_d$. We obtain a lift $E \rightarrow B$ of $\tau^\vee E$ along θ , and thus a map of stable vector bundles $-\tau^\vee E \rightarrow -\theta$ inducing

$$\mathbf{Th}(-\tau^\vee E) \longrightarrow \mathbf{Th}(-\theta).$$

The parametrized Pontrjagin–Thom map associated to the pair $(\pi, \hat{l})$ is the composite

$$\Sigma^\infty X_+ \longrightarrow \mathbf{Th}(-\theta). \tag{3}$$

For a bundle over a point $\pi: W \rightarrow *$, 3 is precisely the Pontrjagin–Thom construction $S \rightarrow \mathbf{Th}(-\theta)$ of an embedding of W in Euclidean space.

1.5. Moduli spaces of manifolds

In this section, we introduce the universal instances of the kind of bundles considered in the previous section. These classifying spaces below are the objects of the main theorems of [GR14; GR18; GR17].

Let W be a closed d -manifold. The topological group $\text{Diff}(W)$ acts on the space of θ -structures $\text{Bun}(TW, \theta^* \gamma_d)$ on the left by precomposing with the differential of a diffeomorphism.

Definition 1.7. Let W be a closed d -dimensional manifold, and $\theta : B \rightarrow BO(d)$ a map. We write

$$\mathcal{M}^\theta(W) = \text{Bun}(TW, \theta^* \gamma_d) // \text{Diff}(W)$$

for the space of homotopy orbits of the action of $\text{Diff}(W)$ ⁷. If $\hat{l}_W$ is a θ -structure on W , we write $\mathcal{M}^\theta(W; \hat{l}_W)$ for the path component of the $\text{Diff}(W)$ -orbit of $\hat{l}_W$.

Example 1.6. Let $\theta = \text{id}$. We write $\mathcal{M}^{\text{id}}(W) = \mathcal{M}(W)$. The space $\text{Bun}(TW, \theta^* \gamma_d)$ is contractible – see the proof of Lemma 1.1 –, and therefore

$$\mathcal{M}(W) \simeq B\text{Diff}(W).$$

Example 1.7. Let $\theta^+ : BSO(d) \rightarrow BO(d)$. We write $\mathcal{M}^{\theta^+}(W) = \mathcal{M}^+(W)$. If W is not orientable, then $\mathcal{M}^+(W) = \emptyset$. If W is orientable, then the space of orientations $\text{Bun}(TW, \gamma_d^+)$ has two path components, each of them weakly contractible. Indeed, pick a classifying map τ_W for TW , and consider the weak equivalence $\text{Map}_{/BO(d)}(\tau_W, \theta^+) \rightarrow \text{Bun}(TW, \gamma^+)$ of Lemma 1.1. The former space is 1-co-connected, that is, its path components are weakly contractible, because θ^+ is a 1-co-connected map. (For any map of spaces $\tau : X \rightarrow Z$ and any n -co-connected map⁸ $\theta : Y \rightarrow Z$, the extension problem $D^{k+1} \leftarrow S^k \rightarrow \text{Map}_{/Z}(\tau, \theta)$ can be solved for all $k \geq n$, by adjunction and obstruction theory.)

Each path component $\mathcal{M}^+(W; \hat{l})$ of $\mathcal{M}^+(W)$ is a model for the classifying space of the group $\text{Diff}^+(W)$ of orientation preserving diffeomorphisms,

$$\mathcal{M}^+(W; \hat{l}) \simeq B\text{Diff}^+(W).$$

Indeed, if $\text{Bun}(TW, \gamma^+)_i$ denotes the path component containing $\hat{l}$, then

$$\mathcal{M}^+(W; \hat{l}) \simeq \text{Bun}(TW, \gamma^+)_i // \text{Diff}^+(W)$$

where $\text{Bun}(TW, \gamma^+)_i$ is contractible – cf. Appendix 4.1.

Therefore, if W admits an orientation reversing diffeomorphism, then $\mathcal{M}^+(W) \simeq B\text{Diff}^+(W)$. If all diffeomorphisms of W are orientation preserving, then $\mathcal{M}^+(W) \simeq B\text{Diff}^+(W) \sqcup B\text{Diff}^+(W)$ and $B\text{Diff}^+(W) = B\text{Diff}(W)$.

In general, $\mathcal{M}^\theta(W)$ classifies pairs (π, l) of a W -bundle $\pi : E \rightarrow X$, and a bundle map $T^v E \rightarrow \theta_d^* \gamma_d$.

1.6. Homology of moduli spaces of manifolds

We state the main theorem [GR17, Corollary 1.9] in the case of orientation, in section 1.6.4. However, it will help us to study more general tangential structures than orientation, and these will arise as Moore–Postnikov factorizations 1.6.1. The main theorem identifies homology of $B\text{Diff}^+(W)$ of a closed, simply connected manifold W of dimension $2n \geq 6$ with that of homotopy theoretic objects $\Omega^\infty \text{MT}\theta$. We define the Thom spectra $\text{MT}\theta$ in section 1.6.2, and then digress from the road to the main theorem in order to examine these objects for latter use. Finally, the range of the identification will depend on the genus of the manifold, defined in section 1.6.3.

⁷See Appendix 4.1 on actions of topological monoids.

⁸Defined in section 1.6.1.

1.6.1 Moore–Postnikov factorizations

We say that a space X is n -connected, for an integer $n \geq -1$, if every map $S^k \rightarrow X$ admits an extension to $D^{k+1} \rightarrow X$ for all $-1 \leq k \leq n$ ⁹. A map $f : X \rightarrow Y$ is called n -connected, for natural number $n \geq 0$, if all of its homotopy fibers are $(n-1)$ -connected.

If X is non-empty, then X is n -connected precisely if $\pi_k(X, x) = 0$ for all $0 \leq k \leq n$ and all basepoints $x \in X$; and f is n -connected precisely if $\pi_k(f) : \pi_k(X, x) \rightarrow \pi_k(Y, f(x))$ is an isomorphism for all $0 \leq k < n$ and an epimorphism for $k = n$.

We say that a space X is n -co-connected if every map $S^k \rightarrow X$ admits an extension to $D^{k+1} \rightarrow X$ for all $n \leq k$. A map $f : X \rightarrow Y$ is n -co-connected if all its homotopy fibers are n -co-connected. If X is non-empty, then this is equivalent to $\pi_k(X, x) = 0$ for all $n \leq k$. If Y is non-empty and $\pi_0(f)$ is surjective, then f is n -co-connected precisely if $\pi_k(f)$ is an isomorphism for all $n \leq k$ and a monomorphism for $k = n$.

For any map $f : A \rightarrow X$ of spaces and any $n \geq 0$ there is a factorization $f = h \circ g : A \rightarrow B \rightarrow X$ with the property that g is n -connected and h is n -co-connected. Such a factorization is called the n th stage of a Moore–Postnikov factorization, and is unique up to weak equivalence under A and over X . (See [Hat01, 4.71].)

Moreover, such a factorization may be obtained with g a cofibration and h a fibration as follows. We inductively construct factorizations $f = h_k \circ g_k : A \rightarrow B^k \rightarrow X$ for $k \geq n$, starting with $B^n = A$. To simplify the exposition, suppose that A is path connected, so we may suppose that $n \geq 1$; let $a \in A$ be a basepoint. Suppose that B^k has been constructed, and for each commutative diagram

$$\begin{array}{ccc} S^k & \longrightarrow & D^{k+1} \\ \downarrow & & \downarrow \\ B^k & \xrightarrow{h_k} & X, \end{array}$$

where $S^k \rightarrow B^k$ either represents a generator for the kernel of $\pi_k(h_k) : \pi_k(B^k, a) \rightarrow \pi_k(X, h_k(a))$ and $D^{k+1} \rightarrow X$ is a nullhomotopy witnessing it, or $S^k \rightarrow B^k$ is the constant map to $a \in A$ and $D^{k+1} \rightarrow S^{k+1} \rightarrow X$ a generator for $\pi_{k+1}(X, h_k(a))$, attach a $(k+1)$ -cell to B^k along $S^k \rightarrow B^k$. We let B^k be the relative CW complex obtained from B^k in this way. The above maps provide an extension h_{k+1} of h_k along the cofibration $B^k \hookrightarrow B^{k+1}$. Let $B = \bigcup_{k \geq n} B^k$.

The inclusion $g : A \hookrightarrow B$ is a cofibration. We may factor $h : B \rightarrow B' \rightarrow X$ as a cofibration $B \rightarrow B'$ which is also a weak equivalence, and a fibration $h' : B' \rightarrow X$. We obtain a cofibration $g' : A \rightarrow B'$ and a fibration $h' : B' \rightarrow X$ with the required properties.

1.6.2 The spectra $\mathbf{MT}\theta$

Definition 1.8. Let $\theta_d : B \rightarrow \mathbf{BO}(d)$ be a map. We define $\mathbf{MT}\theta_d$ as the Thom spectrum $\mathbf{Th}(-\theta_d)$.

Let us unravel the definition – cf. Definition 1.3. Let $\theta_{d,n} : B(\mathbb{R}^{n+d}) \rightarrow \mathbf{Gr}_d(\mathbb{R}^{n+d})$ be the homotopy pullback

$$\begin{array}{ccc} B(\mathbb{R}^{n+d}) & \longrightarrow & B \\ \downarrow \theta_{d,n} & & \downarrow \theta_d \\ \mathbf{Gr}_d(\mathbb{R}^{n+d}) & \longrightarrow & \mathbf{BO}(d). \end{array}$$

⁹We use the convention that $S^{-1} = \emptyset$, so a (-1) -connected space is a non-empty space.

The $(n + d)$ th space of $\text{MT}\theta_d$ is $\text{Th}(\theta_{d,n}^* \gamma_{d,n}^\perp)$, where $\gamma_{d,n}^\perp$ is the n -dimensional bundle orthogonal to the d -dimensional tautological bundle $\gamma_{d,n}$, so that $\gamma_{d,n} \oplus \gamma_{d,n}^\perp$ is the trivial bundle over $\text{Gr}_d(\mathbb{R}^{n+d})$.

Let $\theta : B \rightarrow \text{BO}$ be a map, and for a natural number d consider the homotopy pullback $\theta_d : B(d) \rightarrow \text{BO}(d)$ of θ along $\text{BO}(d) \rightarrow \text{BO}$. We have the following maps of spectra.

(i) A stabilization map

$$s : \Sigma^{-1} \text{MT}\theta_{d-1} \longrightarrow \text{MT}\theta_d$$

which on $(n + d)$ th spaces is given by the map $\text{Th}(\theta_{d-1,n}^* \gamma_{d-1,n}^\perp) \rightarrow \text{Th}(\theta_{d,n}^* \gamma_{d,n}^\perp)$ induced by the embeddings $\text{Gr}_{d-1}(\mathbb{R}^{n+d-1}) \hookrightarrow \text{Gr}_d(\mathbb{R}^{n+d-1} \times \mathbb{R}) = \text{Gr}_d(\mathbb{R}^{n+d})$.

(ii) A map

$$\iota : \text{MT}\theta_d \longrightarrow \Sigma^\infty B(d)_+$$

which on $(n + d)$ th spaces is the composite

$$\text{Th}(\theta_{d,n}^* \gamma_{d,n}^\perp) \longrightarrow \text{Th}(\theta_{d,n}^* \gamma_{d,n}^\perp \oplus \theta_{d,n}^* \gamma_{d,n}) \cong B(\mathbb{R}^{n+d})_+ \wedge S^{n+d} \rightarrow B(d)_+ \wedge S^{n+d}.$$

(iii) The parametrized Pontrjagin–Thom map

$$\partial : \Sigma^\infty B(d)_+ \longrightarrow \text{MT}\theta_{d-1}$$

obtained from the S^{d-1} -bundle $S(\theta_d^* \gamma_d) \rightarrow B(d)$, which models $B(d-1) \rightarrow B(d)$ and thus admits a tautological θ_{d-1} structure. (Note that the sphere bundle is fiberwise embedded $S(\theta_d^* \gamma_d) \subset B(d) \times \mathbb{R}^\infty$.)

Proposition 1.8. *The maps s , ι and ∂ form a cofibration sequence*

$$\Sigma^{-1} \text{MT}\theta_{d-1} \xrightarrow{s} \text{MT}\theta_d \xrightarrow{\iota} \Sigma^\infty B(d)_+ \xrightarrow{\partial} \text{MT}\theta_{d-1}$$

Proof. [Gal+09, Prop. 3.1] and [Gal06, p. 2.1]. □

The stabilization maps s form a system of spectra

$$\text{MT}\theta_0 \longrightarrow \Sigma \text{MT}\theta_1 \longrightarrow \cdots \longrightarrow \Sigma^{d-1} \text{MT}\theta_{d-1} \longrightarrow \Sigma^d \text{MT}\theta_d \longrightarrow \cdots$$

with a map to the Thom spectrum $\text{M}\theta = \text{Th}(\theta)$ of the virtual bundle $\theta : B \rightarrow \text{BO}$, which gives an equivalence on the colimit. The maps $\Sigma^d \text{MT}\theta_d \rightarrow \text{M}\theta$ are described as follows (for $\theta = \text{id}$). The n th space of $\Sigma^d \text{MTO}(d)$ is $\text{Th}(\gamma_{d,n}^\perp)$. The homeomorphism $\text{Gr}_d(\mathbb{R}^{n+d}) \cong \text{Gr}_n(\mathbb{R}^{n+d})$ is covered by a bundle isomorphism $\gamma_{d,n}^\perp \cong \gamma_{n,d}$, and thus we get an homeomorphism of Thom spaces and a map

$$\text{Th}(\gamma_{d,n}^\perp) \cong \text{Th}(\gamma_{n,d}) \longrightarrow \text{Th}(\gamma_{n,\infty}),$$

induced by $\text{Gr}_n(\mathbb{R}^{n+d}) \hookrightarrow \text{BO}(n)$. The target is the n th space of $\text{M}\theta$. The displayed map of Thom spaces is $(d + n)$ -connected, and therefore the induced map of spectra $\Sigma^d \text{MTO}(d) \rightarrow \text{M}\theta$ is d -connected; in the colimit, this gives the equivalence.

Cobordism interpretation. Let $\theta: B \rightarrow BO$ be a fibration and write $\theta_d: B(d) \rightarrow BO(d)$ for the pullback along $BO(d) \rightarrow BO$. The composite

$$-\theta_d: B(d) \longrightarrow BO(d) \longrightarrow BO \xrightarrow{\iota} BO$$

is a virtual vector bundle of rank $-d$, which we may model by a fibration. Then $\Omega_n^{-\theta_d}$ is in bijection with bordism classes of pairs (M^n, l) ,

$$\begin{array}{ccc} & B & \\ l \nearrow & & \downarrow \theta_d \\ M^n & \xrightarrow{\tau_M} & BO \end{array}$$

where $l: M \rightarrow B$ is a lift along θ_d of the stable tangent bundle τ_M .

Then the map

$$s: \Omega_n^{-\theta_d} \cong \pi_n(\Sigma^d MT\theta_d) \longrightarrow \pi_n(M\theta) \cong \Omega_n^\theta \cong \Omega_n^{-\theta}$$

sends a pair $[M, l]$ to the class of M with the structure $l: M \rightarrow B(d) \rightarrow B$.

$$\begin{array}{ccccc} & B(d) & \longrightarrow & B & \\ l \nearrow & \downarrow \theta_d & & \downarrow \theta & \\ M^n & \xrightarrow{\tau_M} & BO & \xlongequal{\quad} & BO. \end{array}$$

Lemma 1.9. Let $\theta: B \rightarrow BSO \rightarrow BO$ be a fibration such that B is path connected. Then

- $\pi_0(\iota): \pi_0(MT\theta_d) \longrightarrow \pi_0(\Sigma^\infty B(d)_+) \cong \mathbb{Z}$ sends a cobordism class $[M, l]$ to the evaluation of the Euler class $\langle e(l^*\theta_d\gamma_d), [M] \rangle$, and
- $\pi_0(\partial): \pi_0(\Sigma^\infty BO(d)_+) \longrightarrow \pi_0(MT\theta_{d-1})$ sends a generator to the cobordism class $[S^{d-1}, l]$ of the fiber of $S(\theta_d^*\gamma_d) \rightarrow B(d)$ with its tautological θ_{d-1} structure on the vertical tangent bundle.

Therefore, the composite is given by multiplication by the Euler characteristic of S^{d-1} .

Proof. The commutative diagram given by pullbacks

$$\begin{array}{ccccc} S^{d-1} \cong SO(d)/SO(d-1) & \longrightarrow & B(d-1) & \longrightarrow & BO(d-1) \\ \downarrow & & \downarrow & & \downarrow \\ * & \longrightarrow & B(d) & \longrightarrow & BO(d) \end{array}$$

maps to BO , so it induces a diagram of Thom spectra

$$\begin{array}{ccc} \Sigma_+^\infty(S^{d-1}) & \longrightarrow & \Sigma^{d-1} MT\theta_{d-1} \\ \downarrow & & \downarrow \\ S & \longrightarrow & \Sigma^d MT\theta_d. \end{array}$$

We get a map on cofibers $\Sigma^{\infty+1}S^{d-1} \rightarrow \Sigma^{\infty+d}B(d)_+$, which is an isomorphism on π_d because $B(d)$ is path connected. Therefore we study the top horizontal arrow

$$\begin{array}{ccc} \pi_0(\mathrm{MT}\theta_d) & \xrightarrow{\pi_0(\iota)} & \pi_{d-1}(S^{d-1}) \cong \mathbb{Z} \\ \downarrow & & \downarrow \\ H_0(\mathrm{MT}\theta_d; \mathbb{Z}) & \longrightarrow & H_{d-1}(S^{d-1}; \mathbb{Z}) \cong \mathbb{Z}. \end{array}$$

Just as in the proof of 2.8, we may identify $\pi_0(\mathrm{MT}\theta_d) \rightarrow \mathbb{Z}$ with the functional

$$\Omega_d^{-\theta_d} \longrightarrow \mathbb{Z}, \quad [M^d, l] \longmapsto \langle e(l^*\theta_d^*\gamma_d), [M] \rangle.$$

Indeed, by definition of the Euler class of the spherical fibration $S^{d-1} \rightarrow S(\theta_d^*\gamma_d) \rightarrow B(d)$ (see Appendix 4.4), the generator of $H^{2n-1}(S^{2n-1}; \mathbb{Z})$ transgresses to $\theta_d^*e \in H^d(B(d); \mathbb{Z})$ in the cohomological Serre spectral sequence for this fibration; here $e \in H^d(\mathrm{BSO}(d); \mathbb{Z})$ is the Euler class. This finishes the proof of the first point.

For the second point, the map ∂ is the parametrized Pontrjagin–Thom map 3 for the sphere bundle $S(\theta_d^*\gamma_d) \rightarrow B(d)$ with its θ_{d-1} -structure. Therefore, on homotopy groups it sends $[x] \in \pi_0(B(d))$ to the Pontrjagin–Thom map of the fiber S_x^{d-1} . (Observe that $[x]$ maps to the generator under $\pi_0(B(d)) \subset \pi_0(B(d)_+) \rightarrow \pi_0(\Sigma^\infty B(d)_+)$.) $\square$

1.6.3 Genus

Let W be a compact path connected manifold of dimension $2n$. We define the *genus* of W to be the natural number

$$g(W) = \max\{g \geq 0 \mid \text{there exists a manifold } M \text{ and a diffeomorphism } W \cong M \sharp (S^n \times S^n)^{\sharp g}\}. \quad (4)$$

An upper bound of the genus is $\lfloor b_n/2 \rfloor$ where b_n is the rank of $H^n(W; \mathbb{Z})$. That is because the quotient map $M \sharp N \rightarrow M \vee N$ induces an isomorphism on homology in degrees $0 < * < 2n-1$ (also in degree $2n-1$ if N is orientable). For better estimates, see [GR20].

1.6.4 Main theorem

Let W be a closed, $2n$ -dimensional, oriented manifold. Pick a Moore–Postnikov n -stage factorization of the map underlying the orientation, that is a factorization

$$W \xrightarrow{l_W} B \xrightarrow{u} \mathrm{BSO}(2n)$$

into an n -connected cofibration l_W and an n -co-connected fibration u . Let

$$\theta : B \xrightarrow{u} \mathrm{BSO}(2n) \longrightarrow \mathrm{BO}(2n).$$

Denote by $\mathrm{hAut}(u)$ the topological monoid consisting of weak equivalences $B \rightarrow B$ commuting with the map u . This group-like topological monoid¹⁰ acts, via pullback, by bundle automorphisms on the vector bundles defining $\mathrm{MT}\theta$. Thus $\mathrm{hAut}(u)$ acts on $\Omega^\infty \mathrm{MT}\theta$. Denote by $g(W)$ the genus of the manifold.

¹⁰See Appendix 4.1 for the definition.

Theorem 1.10 ([GR17, 1.9]). *Let W be a closed, $2n$ -dimensional, oriented manifold. There is a map*

$$B\text{Diff}^+(W) \longrightarrow (\Omega^\infty \text{MT}\theta) // \text{hAut}(u),$$

which, if regarded as a map to the path component that it hits, W is simply connected and $2n \geq 6$, then it induces an isomorphism in integral homology in degrees $$ $\leq (g(W) - 3)/2$.*

Example 1.11. Consider the manifold $W = (S^n \times S^n)^{\sharp g}$, of genus g , equipped with an orientation. Since W is $(n-1)$ -connected, a Moore–Postnikov n -stage consists of $W \rightarrow \text{BO}(2n)\langle n \rangle \rightarrow \text{BSO}(2n)$ where $\text{BO}(2n)\langle n \rangle \rightarrow \text{BO}(2n)$ is the n -connected cover¹¹. We will write $\text{MT}\theta = \text{MO}\langle n \rangle$. Section 3 is a more detailed and elaborate example.

Let us explain how to obtain the map in Theorem 1.10. Let W be a closed $2n$ -manifold, let θ and θ' be two tangential structures, and let u be a fibration which is a map between the tangential structures,

$$\theta : B \xrightarrow{u} B' \xrightarrow{\theta'} \text{BO}(2n).$$

Write $\text{Bun}_n(TW, \theta^*\gamma_{2n})$ for the space of bundle maps $\hat{l}$ whose underlying map $l : W \rightarrow B$ is n -connected. The topological monoid $\text{hAut}(u)$ acts on $\text{Bun}_n(TW, \theta^*\gamma_{2n})$ by postcomposing with the bundle maps $\theta^*\gamma_{2n} \rightarrow \theta'^*\gamma_{2n}$ that its elements induce via pullback.

There is a forgetful map $\hat{u}_* : \text{Bun}_n(TW, \theta^*\gamma_{2n}) \rightarrow \text{Bun}(TW, \theta'^*\gamma_{2n})$ given by $\hat{l} \mapsto \hat{u} \circ \hat{l}$. Every element $f \in \text{hAut}(u)$ satisfies $u = u \circ f$, thus the forgetful map commutes with the action of $\text{hAut}(u)$ on the domain.

Lemma 1.12 ([GR17, 9.2]). *Assume that u is an n -co-connected fibration. Then the induced map*

$$\text{Bun}_n(TW, \theta^*\gamma_{2n}) // \text{hAut}(u) \longrightarrow \text{Bun}(TW, \theta'^*\gamma_{2n}) \quad (5)$$

is a homotopy equivalence onto the path components which it hits.

Addendum 1.13. The assumption that u is a fibration guarantees that u is a fibrant object in the category of spaces over B' , so multiplication $\cdot f : \text{hAut}(u) \rightarrow \text{hAut}(u)$ is a weak equivalence (cf. Lemma 1.1), i.e., the topological monoid is group-like. (Here $B \rightarrow B'$ is cofibrant, which just means that B is cofibrant in spaces, because we may provide W with a CW complex structure via Morse theory, and B may be constructed from W by attaching cells, as shown in section 1.6.1.)

Let $\hat{l}'_W \in \text{Bun}(TW, \theta'^*\gamma_{2n})$. The fiber over $\hat{l}'_W$ of the fibration $\hat{u}_*$ may be identified up to weak equivalence with the space $\text{Map}_{n/B'}(l'_W, u)$ of n -connected lifts

$$\begin{array}{ccc} & & B \\ & \nearrow \hat{l}'_W & \downarrow u \\ W & \xrightarrow{l'_W} & B' \end{array}$$

If this fiber is non-empty, pick a lift l_W . Precomposition with l_W gives a map of mapping spaces $l_W^* : \text{Map}_{/B'}(u, u) \rightarrow \text{Map}_{/B'}(l'_W, u)$ in the category of spaces over B' . The assumption that u is n -co-connected and the condition that l_W is n -connected allow to show that this map is a weak equivalence.

¹¹The n -connected cover of a space X is an n -connected space $X\langle n \rangle$ with an n -co-connected map $X\langle n \rangle \rightarrow X$.

The map restricts to a weak equivalence $\mathrm{hAut}(u) \rightarrow \mathrm{Map}_{n/B'}(l'_W, u)$ onto the fiber F of u_* over l'_W , given by the action $f \mapsto f \circ l_W$. Translating back to bundle maps, $\mathrm{hAut}(u)$ acts fiber-wise on $\mathrm{Bun}_n(TW, \theta^* \gamma_{2n})$ by weak equivalences, thus identifying these fibers with the fibers of $\mathrm{Bun}_n(TW, \theta^* \gamma_{2n}) \rightarrow \mathrm{Bun}_n(TW, \theta^* \gamma_{2n}) // \mathrm{hAut}(u)$. We conclude that the map between base spaces is a weak equivalence by the homotopy long exact sequence and the five lemma.

The topological group $\mathrm{Diff}(W)$ acts on both spaces in 5 by precomposing with the differential of diffeomorphisms, and the map 5 is $\mathrm{Diff}(W)$ -equivariant. Taking homotopy orbits of the action of $\mathrm{Diff}(W)$ we obtain a map

$$\mathcal{M}_n^\theta(W) // \mathrm{hAut}(u) \longrightarrow \mathcal{M}^{\theta'}(W), \quad (6)$$

which is also a homotopy equivalence onto the path components it hits; here, $\mathcal{M}_n^\theta(W)$ denotes the union of those path components $\mathcal{M}^\theta(W, \hat{l}_W)$ for which l_W is n -connected.

Let $\hat{l}'_W$ be a θ' -structure. Suppose that the fiber of $\hat{u}_*$ over $\hat{l}'_W$ is non-empty, for example if u is an n -co-connected fibration arising from the Moore–Postnikov factorization of l'_W . The path components of the $\mathrm{Diff}(W)$ -orbits of $\hat{l}_W$ and $\hat{l}'_W$ are what we denoted by $\mathcal{M}^\theta(W; \hat{l}_W)$ and $\mathcal{M}^{\theta'}(W; \hat{l}'_W)$, respectively. Then 6 restricts to a homotopy equivalence

$$\mathcal{M}^\theta(W; \hat{l}_W) // G_{\hat{l}_W} \longrightarrow \mathcal{M}^{\theta'}(W; \hat{l}'_W) \quad (7)$$

where $G_{\hat{l}_W} \leq \mathrm{hAut}(u)$ is the submonoid consisting of those path components whose action on $\mathcal{M}_n^\theta(W)$ restricts to $\mathcal{M}^\theta(W; \hat{l}_W) \subset \mathcal{M}_n^\theta(W)$. That is, $f \in \mathrm{hAut}(u)$ is an element of $G_{\hat{l}_W}$ if there exists a diffeomorphism of W such that $\hat{f} \circ \hat{l}_W \circ d\phi$ is in the same path component of the space of bundle maps as $\hat{l}_W$. We say that f preserves the θ -structure $\hat{l}_W$ up to diffeomorphism. Note that such a diffeomorphism ϕ necessarily preserves the θ' -structure $\hat{l}'_W$, because $\hat{l}'_W = \hat{u} \circ \hat{l}_W$.

The parametrized Pontrjagin–Thom map $\alpha: \mathcal{M}^\theta(W, \hat{l}_W) \rightarrow \Omega^\infty \mathrm{MT}\theta$ commutes with the actions of $\mathrm{hAut}(u)$ and thus induces a map

$$\alpha // \mathrm{hAut}(u): \mathcal{M}^{\theta'}(W, \hat{l}'_W) \longrightarrow (\Omega^\infty \mathrm{MT}\theta) // \mathrm{hAut}(u).$$

This map is the one in the statement of the Theorem 1.10.

1.6.5 A fiber sequence

Using the homotopy equivalence 7 we obtain a fiber sequence

$$\mathcal{M}^\theta(W, \hat{l}_W) \longrightarrow \mathcal{M}^{\theta'}(W, \hat{l}'_W) \longrightarrow B(G_{\hat{l}_W}). \quad (8)$$

We have a map of fiber sequences

$$\begin{array}{ccccc} \mathcal{M}^\theta(W, \hat{l}_W) & \longrightarrow & \mathcal{M}^{\theta'}(W, \hat{l}'_W) & \longrightarrow & B(G_{\hat{l}_W}) \\ \downarrow \alpha & & \downarrow \alpha // \mathrm{hAut}(u) & & \downarrow \\ \Omega^\infty \mathrm{MT}\theta & \longrightarrow & (\Omega^\infty \mathrm{MT}\theta) // (\mathrm{hAut}(u)) & \longrightarrow & B(\mathrm{hAut}(u)), \end{array}$$

in which the first two vertical maps are homology equivalences in the situation and range of degrees of Theorem 1.10. In practice, one may be interested in the homology or cohomology of the upper middle term, so that of the outer terms is computed and inputted into the Serre spectral sequence of the fiber sequence.

2. ABELIAN QUOTIENTS OF MAPPING CLASS GROUPS OF HIGHLY CONNECTED MANIFOLDS

We follow [GR16] to compute the abelianizations of the mapping class groups of the manifolds

$$W_g^{2n} = \#^g(S^n \times S^n)$$

obtained by a g -fold connected sum for $n \geq 3$ and $g \geq 5$. Choose and fix a disc $D^{2n} \subset S^n \times S^n$. Let $\text{Diff}^+(W_g^{2n})$ be the topological group of orientation preserving diffeomorphisms of W_g^{2n} , and let $\text{Diff}(W_g^{2n}, D^{2n})$ be the subgroup of those diffeomorphisms which fix pointwise an open neighborhood of the disc D^{2n} . Define the *mapping class groups*

$$\Gamma_g^n = \pi_0(\text{Diff}^+(W_g^{2n})), \quad \Gamma_{g,1}^n = \pi_0(\text{Diff}(W_g^{2n}, D^{2n})).$$

Lemma 2.1 ([Kre79]). *The forgetful map $\Gamma_{g,1}^n \rightarrow \Gamma_g^n$ is an isomorphism if $n \geq 3$.*

Given an $(n-1)$ -connected $2n$ -manifold, such as W_g^{2n} , one can associate to it a certain quadratic form $Q_{W_g^{2n}}$ whose underlying bilinear form is the intersection form on $H_n(W_g^{2n}; \mathbb{Z})$ – see [Wal62]. The action of orientation preserving diffeomorphisms on this quadratic form gives a group homomorphism $\Gamma_g^n \rightarrow \text{Aut}(Q_{W_g^{2n}})$. Composing with the Hurewicz map to the first homology group, i.e., the abelianization, we get maps

$$f : \Gamma_{g,1}^n \longrightarrow \Gamma_g^n \longrightarrow H_1(B \text{Aut}(Q_{W_g^{2n}}); \mathbb{Z}).$$

If G is a group-like topological monoid, then the map $G \rightarrow \pi_0(G)$ induces a homomorphism $\pi_1(B\pi_0(G)) \rightarrow \pi_1(BG)$, which is an isomorphism because both groups admit a canonical isomorphism from $\pi_0(\pi_0(G))$ and $\pi_0(G)$, respectively, and these are also canonically isomorphic. Via the Hurewicz theorem, we identify their abelianizations. Applying this to $G = \text{Diff}(W_g^{2n}, D^{2n})$ we obtain

$$H_1(B\text{Diff}(W_g^{2n}, D^{2n}); \mathbb{Z}) \xrightarrow{\cong} H_1(B\Gamma_{g,1}^n; \mathbb{Z}). \quad (9)$$

For $g \geq 5$ and $n \geq 3$, a version of Theorem 1.10 for manifolds with boundary [GR17, 1.8] applies in degree 1 to the map

$$B\text{Diff}(W_g^{2n}, D^{2n}) \longrightarrow \Omega^\infty \text{MT}\theta_{2n},$$

where $\theta_{2n} : BO(2n)\langle n \rangle \rightarrow BO(2n)$ denotes the n -connected cover (cf. Example 1.11). Thus

$$H_1(B\text{Diff}(W_g^{2n}, D^{2n}); \mathbb{Z}) \xrightarrow{\cong} H_1(\Omega_0^\infty \text{MT}\theta_{2n}; \mathbb{Z}) \cong \pi_1(\Omega_0^\infty \text{MT}\theta_{2n}) \cong \pi_1(\text{MT}\theta_{2n}),$$

because all path components of $\Omega^\infty \text{MT}\theta_{2n}$ are homotopy equivalent, and by the Hurewicz theorem. We can combine this with 9 to identify

$$\pi_1(\text{MT}\theta_{2n}) \cong H_1(B\Gamma_{g,1}^n; \mathbb{Z}).$$

We denote by $\text{MO}\langle n \rangle$ for the Thom spectrum of the n -connected cover $\theta : BO\langle n \rangle \rightarrow BO$, and by $\Omega_{2n+1}^{(n)}$ the cobordism groups of closed $(2n+1)$ -dimensional manifolds together with a $BO\langle n \rangle \rightarrow BO$

structure on their stable normal bundle. The stabilization map (i) $s: \Sigma^{2n} \text{MT}\theta_{2n} \rightarrow \text{MO}\langle n \rangle$ induces a group homomorphism

$$\pi_1(s): \pi_1(\text{MT}\theta_{2n}) \longrightarrow \pi_{2n+1}(\text{MO}\langle n \rangle) \cong \Omega_{2n+1}^{(n)}.$$

The following is only a part of the main theorem in [GR16].

Theorem 2.2 ([GR16]). *For all natural numbers $n \geq 3$ and $g \geq 0$ the map*

$$\pi_1(s) \oplus f: H_1(B\Gamma_{g,1}^n; \mathbb{Z}) \longrightarrow \Omega_{2n+1}^{(n)} \oplus H_1(B \text{Aut}(Q_{W_g^{2n}}); \mathbb{Z})$$

is surjective. For $g \geq 5$ it is an isomorphism, and, furthermore,

$$H_1(B \text{Aut}(Q_{W_g^{2n}}); \mathbb{Z}) \cong \begin{cases} (\mathbb{Z}/2)^2 & n \text{ even,} \\ 0 & n = 1, 3, \text{ or } 7, \\ \mathbb{Z}/4 & n \text{ otherwise.} \end{cases}$$

We will focus solely on the homomorphism $\pi_1(s)$; that's where the theory on homology of moduli spaces of manifolds is involved.

2.1. A long exact sequence in stable homotopy

We study $\pi_1(s)$ as part of the long exact sequence of the fiber sequence

$$F_{2n} \longrightarrow \text{MT}\theta_{2n} \xrightarrow{s} \Sigma^{-2n} \text{MO}\langle n \rangle.$$

The map s is induced by a map of stable vector bundles as displayed in the right square in 10; this square is a homotopy pullback, therefore we identify the homotopy fibers of the vertical maps, that of $\text{BO}(2n) \rightarrow \text{BO}$ being $\text{SO} / \text{SO}(2n)$ ¹². We have a commutative diagram

$$\begin{array}{ccccc} \text{SO} / \text{SO}(2n) & \longrightarrow & \text{BO}(2n)\langle n \rangle & \longrightarrow & \text{BO}(2n) \\ \downarrow & & \downarrow & & \downarrow \\ * & \longrightarrow & \text{BO}\langle n \rangle & \longrightarrow & \text{BO}. \end{array} \tag{10}$$

The whole diagram maps to BO , so it induces maps of Thom spectra

$$\begin{array}{ccc} \Sigma_+^\infty(\text{SO} / \text{SO}(2n)) & \longrightarrow & \Sigma^{2n} \text{MT}\theta_{2n} \\ \downarrow & & \downarrow \\ S & \longrightarrow & \text{MO}\langle n \rangle. \end{array}$$

We get an induced map on cofibers

$$\Sigma^{\infty+1}(\text{SO} / \text{SO}(2n)) \longrightarrow \Sigma^{2n+1} F_{2n},$$

¹²Cf. Appendix 4.4.

which we next show to be $(3n + 1)$ -connected. We can check connectivity on homology by the Hurewicz theorem. We do so for the bottom map in the commutative diagram below given by the relative Thom isomorphism.

$$\begin{array}{ccc} H^*(\Sigma^{\infty+1}(\mathrm{SO} / \mathrm{SO}(2n)); \mathbb{Z}) & \longleftarrow & H^*(\Sigma^{2n+1}F_{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow \\ H^*(\Sigma(\mathrm{SO} / \mathrm{SO}(2n)), *; \mathbb{Z}) & \longleftarrow & H^*(\mathrm{BO}\langle n \rangle, \mathrm{BO}(2n)\langle n \rangle; \mathbb{Z}). \end{array}$$

The bottom map is induced by the map of cofibers $\Sigma(\mathrm{SO} / \mathrm{SO}(2n)) \rightarrow \mathrm{BO}\langle n \rangle / \mathrm{BO}(2n)\langle n \rangle$. Now, from the Serre spectral sequence for the fiber sequence $\mathrm{SO} / \mathrm{SO}(2n) \rightarrow \mathrm{BO}(2n)\langle n \rangle \rightarrow \mathrm{BO}\langle n \rangle$ we extract the Serre exact sequence, which maps to the long exact sequence of the pair

$$\begin{array}{ccccccc} H_{3n}(\mathrm{SO} / \mathrm{SO}(2n)) & \longrightarrow & H_{3n}(\mathrm{BO}(2n)\langle n \rangle) & \longrightarrow & H_{3n}(\mathrm{BO}\langle n \rangle) & \longrightarrow & H_{3n-1}(\mathrm{SO} / \mathrm{SO}(2n)) \cdots \\ \downarrow & & \parallel & & \parallel & & \downarrow \\ H_{3n+1}(\mathrm{BO}\langle n \rangle, \mathrm{BO}(2n)\langle n \rangle) & \longrightarrow & H_{3n}(\mathrm{BO}(2n)\langle n \rangle) & \longrightarrow & H_{3n}(\mathrm{BO}\langle n \rangle) & \longrightarrow & H_{3n}(\mathrm{BO}\langle n \rangle, \mathrm{BO}(2n)\langle n \rangle) \cdots, \end{array} \quad (11)$$

ending at $H_1(\mathrm{BO}(2n)\langle n \rangle)$. (All coefficients are integral.) We deduce the claim by the 5-lemma.

Remark 2.3. If we consider the fiber sequence $F \rightarrow \mathrm{MTSpin}^c(6) \rightarrow \Sigma^{-6}\mathrm{MSpin}^c$, the same argument shows that the map $\Sigma^{\infty-6}(\mathrm{SO} / \mathrm{SO}(6)) \rightarrow F$ is 1-connected. Indeed, 10 factors as

$$\begin{array}{ccccccc} \mathrm{SO} / \mathrm{SO}(6) & \longrightarrow & B\mathrm{Spin}(6) & \longrightarrow & B\mathrm{Spin}^c(6) & \longrightarrow & \mathrm{BO}(6) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ * & \longrightarrow & B\mathrm{Spin} & \longrightarrow & B\mathrm{Spin}^c & \longrightarrow & \mathrm{BO}. \end{array}$$

The mentioned map is 1-connected, because $B\mathrm{Spin}^c$ is 1-connected so we can only assure that the Serre exact sequence starts at $H_7(\mathrm{SO} / \mathrm{SO}(2n))$.

Now, the long exact sequence looks like

$$\begin{array}{ccccccc} & & \cdots & \longrightarrow & \pi_{2n+2} \mathrm{MO}\langle n \rangle & & \\ & & \searrow \partial_* & & \swarrow & & \\ \pi_{2n+1}^s(\mathrm{SO} / \mathrm{SO}(2n)) & \longrightarrow & \pi_1 \mathrm{MT}\theta_{2n} & \xrightarrow{\pi_1(s)} & \pi_{2n+1} \mathrm{MO}\langle n \rangle & & \\ & & \searrow & & \swarrow & & \\ \pi_{2n}^s(\mathrm{SO} / \mathrm{SO}(2n)) & \longrightarrow & \pi_0 \mathrm{MT}\theta_{2n} & \longrightarrow & \pi_{2n} \mathrm{MO}\langle n \rangle & & \end{array}$$

Since $\mathrm{SO} / \mathrm{SO}(2n)$ is $(2n-1)$ -connected, the Freudenthal's theorem asserts that the stabilization homomorphism $\pi_k(\mathrm{SO} / \mathrm{SO}(2n)) \rightarrow \pi_k^s(\mathrm{SO} / \mathrm{SO}(2n))$ is an isomorphism for $k \leq 2(2n-1)$. This applies for $k = 2n, 2n+1, 2n+2$ as long as $n \geq 2$. Since the map from the Stiefel manifold $V_{2n+m,m} = \mathrm{SO}(2n+m) / \mathrm{SO}(2n) \rightarrow \mathrm{SO} / \mathrm{SO}(2n)$ is $(2n+m-1)$ -connected¹³, the homotopy groups appearing in the long exact sequence are those of Stiefel manifolds $V_{2n+m,m}$ for large enough m . These were computed by Paechter.

¹³See Appendix 4.4.

Lemma 2.4 ([Pae56]). Let $n \geq 2$ be a natural number. Then $\pi_{2n}^s(\text{SO} / \text{SO}(2n)) \cong \mathbb{Z}$, and

$$\pi_{2n+1}^s(\text{SO} / \text{SO}(2n)) \cong \begin{cases} (\mathbb{Z}/2)^2 & n \text{ even,} \\ \mathbb{Z}/4 & n \text{ odd.} \end{cases}$$

Lemma 2.5. The map $\pi_{2n}^s(\text{SO} / \text{SO}(2n)) \rightarrow \pi_0(\text{MT}\theta_{2n})$ is injective. Consequently

$$\pi_1(s): \pi_1(\text{MT}\theta_{2n}) \longrightarrow \pi_{2n+1}(\text{MO}\langle n \rangle)$$

is surjective.

Proof. We show that there is a class in the image of $H^0(\text{MT}\theta_{2n}; \mathbb{Z}) \rightarrow H^{2n}(\Sigma^\infty(\text{SO} / \text{SO}(2n)); \mathbb{Z})$ which is a multiple of a generator of $H^{2n}(\text{SO} / \text{SO}(2n)) \cong \mathbb{Z}$, so the map it induces on π_0 ,

$$\pi_{2n}^s(\text{SO} / \text{SO}(2n)) \longrightarrow \pi_0(\text{MT}\theta_{2n}) \longrightarrow \pi_0(H\mathbb{Z}) = \mathbb{Z},$$

is multiplication by the corresponding integer, by the argument given at the end of section 3.4. The result follows. (The isomorphism $H^{2n}(\text{SO} / \text{SO}(2n); \mathbb{Z}) \cong \mathbb{Z}$ is due to the Hurewicz theorem, and the universal coefficient theorem.)

Naturality of the Thom isomorphism gives commutativity of the diagram

$$\begin{array}{ccc} H^0(\text{MT}\theta_{2n}; \mathbb{Z}) & \longrightarrow & H^{2n}(\Sigma^\infty(\text{SO} / \text{SO}(2n)); \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow \\ H^{2n}(\text{BO}(2n)\langle n \rangle; \mathbb{Z}) & \longrightarrow & H^{2n}(\text{SO} / \text{SO}(2n), *, \mathbb{Z}). \end{array}$$

We reduce to studying the bottom horizontal map.

We shall identify the pullback of the Euler class $e \in H^{2n}(\text{BSO}(2n); \mathbb{Z})$ along the composite $\text{SO} / \text{SO}(2n) \rightarrow \text{BO}(2n)\langle n \rangle \rightarrow \text{BSO}(2n)$ as a class with the desired property. To do so, we may as well pull-it back along the map $S^{2n} \cong \text{SO}(2n+1) / \text{SO}(2n) \rightarrow \text{SO} / \text{SO}(2n) \rightarrow \text{BSO}(2n)$ classifying the oriented tangent bundle of the sphere. Indeed, the first map is $2n$ -connected so it induces an injection on the $2n$ th cohomology.

The image of e under $\tau_{S^{2n}}^*: H^{2n}(\text{BSO}(2n); \mathbb{Z}) \rightarrow H^{2n}(\text{SO} / \text{SO}(2n); \mathbb{Z}) \hookrightarrow H^{2n}(S^{2n}; \mathbb{Z})$ is the Euler class $e(S^{2n})$ of S^{2n} , which, under the Poincaré duality isomorphism $\langle -, [S^{2n}] \rangle: H^{2n}(S^{2n}; \mathbb{Z}) \rightarrow H_0(S^{2n}; \mathbb{Z}) \cong \mathbb{Z}$, maps to the Euler characteristic of S^{2n} by [MS16, 11.12], that is 2. Therefore, the pullback of e to $\text{SO} / \text{SO}(2n)$ is at most divisible by 2. It is divisible by 2 because $e(S^{2n})$ reduces modulo 2 to $w_{2n}(S^{2n}) = 0$, so it must also be zero modulo 2 in $H^{2n}(\text{SO} / \text{SO}(2n); \mathbb{Z})$. $\square$

Remark 2.6. We can rewrite this proof using Lemma 1.9.

Lemma 2.7. The map $\pi_1(s): \pi_1(\text{MTSpin}^c(6)) \rightarrow \pi_7(\text{MSpin}^c)$ is surjective.

Proof. The proof of Lemma 2.8 also works in this case because the map $\text{BO}(6)\langle 3 \rangle = \text{BSpin}(6) \rightarrow \text{BSO}(6)$ factors through $\text{BSpin}^c(6)$. $\square$

Lemma 2.8. If $n = 3, 7$, then ∂_* is surjective. Consequently,

$$\pi_1(s): \pi_1(\text{MT}\theta_{2n}) \longrightarrow \pi_{2n+1}(\text{MO}\langle n \rangle)$$

is injective.

Proof. We prove surjectivity of the top horizontal map in the diagram

$$\begin{array}{ccc} \pi_{2n+2}(\mathrm{MO}\langle n \rangle) & \xrightarrow{\partial_*} & \pi_{2n+1}^s(\mathrm{SO} / \mathrm{SO}(2n)) \cong \mathbb{Z}/4 \\ \downarrow h & & \downarrow h \\ H_{2n+2}(\mathrm{MO}\langle n \rangle; \mathbb{Z}/2) & \longrightarrow & H_{2n+1}(\mathrm{SO} / \mathrm{SO}(2n); \mathbb{Z}/2) \cong \mathbb{Z}/2. \end{array}$$

We denote by h the composition of the Hurewicz homomorphism and the reduction $\mathbb{Z}/2$ coefficients. We prove that the bottom right cohomology group is 1-dimensional, and that the right vertical map and the composite are both surjective. From this we can deduce that ∂_* must hit the units and thus be surjective.

Surjectivity of h . Since $\mathrm{SO} / \mathrm{SO}(2n)$ is $(2n-1)$ -connected, the Hurewicz homomorphism onto the $(2n+1)$ th homology group with integer coefficients is surjective. Moreover, the reduction to $\mathbb{Z}/2$ coefficients $H_{2n+1}(\mathrm{SO} / \mathrm{SO}(2n); \mathbb{Z}) \rightarrow H_{2n+1}(\mathrm{SO} / \mathrm{SO}(2n); \mathbb{Z}/2)$ is also surjective. Indeed, the connecting map to $H_{2n}(\mathrm{SO} / \mathrm{SO}(2n); \mathbb{Z}) \cong \pi_{2n}(\mathrm{SO} / \mathrm{SO}(2n)) \cong \mathbb{Z}$ in the Bockstein exact sequence is zero because multiplication by 2 on this group is an injection – this isomorphism is given by the Hurewicz theorem.

Claim. The map

$$\partial_* : H^{2n+1}(\mathrm{SO} / \mathrm{SO}(2n); \mathbb{Z}) \longrightarrow H^0(\mathrm{MO}\langle n \rangle; \mathbb{Z}/2)$$

sends a generator x_{2n+1} to $(\theta^* w_{2n+2}) \cdot u$, where $w_{2n+2} \in H^{2n+2}(\mathrm{BSO}; \mathbb{Z}/2)$ is the Stiefel–Whitney class, and $u \in H^0(\mathrm{MO}\langle n \rangle; \mathbb{Z}/2)$ the Thom class.

Surjectivity of $\partial_ \circ h$.* The last isomorphism in the diagram is given by the adjoint to the evaluation pairing $\langle x_{2n+1}, - \rangle : H_{2n+1}(\mathrm{SO} / \mathrm{SO}(2n); \mathbb{Z}/2) \rightarrow \mathbb{Z}/2$, so the whole composite is

$$\langle x_{2n+1}, \partial_* \circ h(-) \rangle = \langle (\theta^* w_{2n+2}) \cdot u, h(-) \rangle : \pi_{2n+2}(\mathrm{MO}\langle n \rangle) \longrightarrow \mathbb{Z}/2.$$

Let $[W^{2n+2}, l] \in \Omega_{2n+2}^{(n)}$ with stable tangent bundle τ_W , and write $[W]$ for the fundamental class of the manifold. We know that $h([W^{2n+2}, l])$ maps to $l_*([W])$ under the $\mathbb{Z}/2$ -dual of the Thom isomorphism (cf. [MS16, 11-C] and Remark 4.3). Thus,

$$\langle (\theta^* w_{2n+2}) \cdot u, h(-) \rangle = \langle \theta^* w_{2n+2}, l_*([W]) \rangle = \langle l^* \theta^* w_{2n+2}, [W] \rangle.$$

Observe that $l^* \theta^* w_{2n+2} = \tau_W^* w_{2n+2} = w_{2n+2}(TW)$.

By [MS16, 11.12], $\langle w_{2n+2}(TW), [W] \rangle$ equals the Euler characteristic of W modulo 2. The manifolds $[\mathbb{HP}^2] \in \Omega_8^{(3)}$ and $[\mathbb{OP}^2] \in \Omega_{16}^{(7)}$ (with their unique structures, due to their connectivity) have Euler characteristic 3, and therefore witness that $\Omega_{2n+2}^{(n)} \rightarrow \mathbb{Z}/2$ is surjective. To determine their Euler characteristic, we use that they admit a cell structure with one cell in each dimension which is a multiple of $n+1$, up to $2(n+1)$, for $n=3, 7$ respectively. That is, three cells, all in even dimensions.

Proof of the claim. In what follows, cohomology is taken with coefficients in $\mathbb{Z}/2$. We show that $H^{2n+1}(\mathrm{SO} / \mathrm{SO}(2n))$ is 1-dimensional and that its generator x_{2n+1} transgresses to $w_{2n+2} \in H^{2n+2}(\mathrm{BSO})$ in the cohomological Serre spectral sequence for the fiber sequence

$$\mathrm{SO} / \mathrm{SO}(2n) \longrightarrow \mathrm{BSO}(2n) \longrightarrow \mathrm{BSO}.$$

Assuming that this is true, x_{2n+1} transgresses to $\theta^* w_{2n+2}$ in the spectral sequence for

$$\mathrm{SO} / \mathrm{SO}(2n) \rightarrow \mathrm{BO}(2n)\langle n \rangle \rightarrow \mathrm{BO}\langle n \rangle,$$

because this fiber sequence maps to the one above. Therefore $\partial^*(x_{2n+1}) = (\theta^* w_{2n+2}) \cdot u$, by commutativity of the further right square in 11 (for cohomology) and the Thom isomorphism.

Consider the map of fiber sequences

$$\begin{array}{ccccc} \mathrm{SO} / \mathrm{SO}(2n) & \longrightarrow & \mathrm{BSO}(2n) & \longrightarrow & \mathrm{BSO} \\ \downarrow & & \downarrow & & \parallel \\ \mathrm{SO} / \mathrm{SO}(2n+1) & \longrightarrow & \mathrm{BSO}(2n+1) & \longrightarrow & \mathrm{BSO}. \end{array}$$

By naturality of the Serre spectral sequence, it suffices to prove the claim about the transgression for the bottom fiber sequence. The generator of $H^{2n+1}(\mathrm{SO} / \mathrm{SO}(2n+1))$ transgresses to $w_{2n+2} \in H^{2n+2}(\mathrm{BSO})$ by definition of the Stiefel–Whitney classes – cf. Appendix 4.4.

Regarding the claim about the dimension, by [MS16, 12.4] the $\mathbb{Z}/2$ -cohomology of BSO is the polynomial algebra on the Stiefel–Whitney classes $w_i \in H^i(\mathrm{BSO})$ with $i \geq 2$, and the map $H^*(\mathrm{BSO}) \rightarrow H^*(\mathrm{BSO}(2n))$ is the quotient by the ideal generated by those classes w_i with $i > 2n$. We deduce that $H^{2n+1}(\mathrm{SO} / \mathrm{SO}(2n))$ is 1-dimensional from the long exact sequence 11. $\square$

Lemma 2.9. *The map $\pi_1(s): \pi_1(\mathrm{MTSpin}^c(6)) \rightarrow \pi_7(\mathrm{MSpin}^c)$ is injective.*

Proof. The diagram of Remark 2.3 induces a map of fiber sequences

$$\begin{array}{ccccc} F_6 & \longrightarrow & \mathrm{MTSpin}(6) & \longrightarrow & \Sigma^{-6} \mathrm{MSpin} \\ \downarrow & & \downarrow & & \downarrow \\ F & \longrightarrow & \mathrm{MTSpin}^c(6) & \longrightarrow & \Sigma^{-6} \mathrm{MSpin}^c. \end{array}$$

We get a map of (desuspended) cofibers $\Sigma^{\infty-6} \mathrm{SO} / \mathrm{SO}(6) \rightarrow F_6 \rightarrow F$. The map to F_6 is 3-connected, and the composite is 1-connected, therefore a surjection on π_1 – cf. Remark 2.3. Because of this and Lemma 2.8, we have the following surjections.

$$\begin{array}{ccc} \pi_8(\mathrm{MSpin}) & \xrightarrow{\partial_*} & \pi_1(F_3) \\ \downarrow & & \downarrow \\ \pi_8(\mathrm{MSpin}^c) & \xrightarrow{\partial_*} & \pi_1(F). \end{array}$$

We conclude that $\pi_1(\mathrm{MTSpin}^c(6)) \rightarrow \pi_1(\mathrm{MSpin}^c)$ is injective. $\square$

Using Lemmas 2.4, 2.8, 2.5, and 2.10 below, one can complete a major part of the proof of Theorem 2.2. We refer to [GR16] for the remaining details.

Lemma 2.10 ([GR16, 5.2]). *If $n \neq 3, 7$, then the maps*

$$0 \longrightarrow \pi_{2n+1}^s(\mathrm{SO} / \mathrm{SO}(2n)) \longrightarrow \pi_1 \mathrm{MT}\theta_{2n} \xrightarrow{\pi_1(s)} \pi_{2n+1} \mathrm{MO}\langle n \rangle \longrightarrow 0$$

form a split exact sequence.

2.2. Cobordism classes of spheres

From a computational point of view, we briefly indicate the relationship between the cobordism groups appearing in Theorem 2.2, and the stable homotopy groups of spheres (equivalently, framed cobordism Ω_*^{fr} , by the Pontrjagin–Thom theorem 1.2).

There is a forgetful homomorphism

$$\Omega_{2n+1}^{\text{fr}} \longrightarrow \Omega_{2n+1}^{(n)}$$

which vanishes on the bordism classes of spheres S^k of dimension $k \geq n$. Indeed, any $BO\langle n \rangle \rightarrow BO$ structure l on S^k can be extended to the disk D^{k+1}

$$\begin{array}{ccc} S^k & \xrightarrow{l} & BO\langle n \rangle \\ \downarrow & \nearrow & \downarrow \\ D^{k+1} & \longrightarrow & BO \end{array}$$

if $k \geq n$, because $BO\langle n \rangle \rightarrow BO$ is n -co-connected, so there are no obstructions to doing so (see obstruction theory 3.6.1).

The J -homomorphism $\pi_k(\mathcal{O}) \rightarrow \Omega_k^{\text{fr}}$ sends the homotopy class of a map $\alpha: S^k \rightarrow \mathcal{O}$ to the cobordism class of S^k with the framing obtained from its standard framing $S^k \rightarrow \mathcal{EO}$ and changing it by letting α act fiberwise on $\mathcal{EO} \rightarrow BO$. As observed above, any class in the image of J vanished in $\Omega_{2n+1}^{(n)}$. We obtain a map

$$\rho: \text{coker}(J)_{2n+1} \longrightarrow \Omega_{2n+1}^{(n)}$$

from the cokernel of the J -homomorphism. After work of many authors, [BS20, 1.4] have solved the problem of determining when ρ is an isomorphism.

Theorem 2.11. *Suppose that $n > 1$ and $n \neq 8, 11$. Then*

$$\rho: \text{coker}(J)_{2n+1} \longrightarrow \Omega_{2n+1}^{(n)}$$

is an isomorphism. There are elements $\eta\eta_4 \in \text{coker}(J)_{17}$ and $\eta^3\bar{\kappa} \in \text{coker}(J)_{23}$ such that

$$\text{coker}(J)_{17}/\eta\eta_4 \longrightarrow \Omega_{2n+1}^{(n)} \quad \text{and} \quad \text{coker}(J)_{23}/\eta^3\bar{\kappa} \longrightarrow \Omega_{2n+1}^{(n)}$$

are isomorphisms.

3. ABELIANIZATIONS OF MAPPING CLASS GROUPS OF HYPERSURFACES IN $\mathbb{CP}^4$

An algebraic hypersurface $V_d \subset \mathbb{CP}^r$ of degree d is the set of zeroes of a homogeneous complex polynomial $p \in \mathbb{C}[x_0, \dots, x_r]$ of degree d . If $\gamma_{1,r} \rightarrow \mathbb{CP}^r$ denotes the tautological complex vector bundle, then p specifies a holomorphic section s_p of the holomorphic complex vector bundle $\mathcal{O}(d) = \text{Hom}_{\mathbb{C}}((\gamma_{1,r})^{\otimes d}, \underline{\mathbb{C}}) \rightarrow \mathbb{CP}^r$ given by taking linear dual fiber-wise; the preimage under s_p of the zero section is V_d . If s_p is transverse to the zero section, then V_d is a smooth complex submanifold of complex dimension $r - 1$. The diffeomorphism type of V_d depends only on the degree d .

A complex manifold admits a canonical orientation, because the underlying real vector space of a complex vector space does. Therefore, in what follows, this choice of orientation is implicit. From now on $r = 4$, and for example $V_1 \cong \mathbb{CP}^3$.

This family of manifolds was considered in [GR20, §5.3]. By taking the same approach as in [GR16], of using their Theorem 1.10, and incorporating our own calculations in section 3.5 and one of $\Omega_7^{\text{Spin}^c}$ by [BG87], we obtained the following result in terms of the genus $g(V_d)$. The genus of these manifolds is computed in section 3.3 in terms of the degree d . Recall that a group is called *perfect* if its abelianization is trivial.

Proposition 3.1. *Let $d \geq 1$ be an integer, and $V_d \subset \mathbb{CP}^4$ be a smooth hypersurface of degree d . Let $g \geq \max\{5 - g(V_d), 0\}$. Then the mapping class group $\pi_0(\text{Diff}^+(V_d \sharp (S^3 \times S^3)^{\sharp g}))$ is perfect.*

3.1. Strategy

From now on, we fix a $g \geq \max\{5 - g(V_d), 0\}$, we write $W_d = V_d \sharp (S^3 \times S^3)^{\sharp g}$, and we fix $\hat{l}_{W_d}^+$ the canonical orientation on W_d . As in section 2, we consider the map $\text{Diff}^+(W_d) \rightarrow \pi_0(\text{Diff}^+(W_d))$, which induces an isomorphism on the first homology groups, and our purpose is then to calculate

$$H_1(B\text{Diff}^+(W_d); \mathbb{Z}).$$

We will use Theorem 1.10 to calculate this group. (Our assumption on g is solely to ensure that the theorem applies in degree 1.) For this, we take $\theta^+ : BSO(6) \rightarrow BO(6)$ and we start by determining the Moore–Postnikov 3-stage factorization $u^+ \circ \hat{l}_{W_d}^+ : W_d \rightarrow B_d \rightarrow BSO(6)$ of the orientation $\hat{l}_{W_d}^+ : W_d \rightarrow BSO(6)$. This will depend on the parity of d , and we write $\theta_d : B_d \rightarrow BO(6)$ for the obtained map.

Let $G \leq \text{hAut}(u^+)$ be the submonoid of those path components that preserve the θ_d -structure $\hat{l}_{W_d}^+$ up to orientation preserving diffeomorphism of W_d – the map u^+ and these monoids also depend on d , but we omit this in the notation. The approach is to use the fiber sequence 8, in our case

$$\mathcal{M}^{\theta_d}(W_d; \hat{l}_{W_d}^+) \longrightarrow \mathcal{M}^+(W_d; \hat{l}_{W_d}^+) \longrightarrow BG.$$

where $\mathcal{M}^+(W_d, \hat{l}_{W_d}^+) \simeq B\text{Diff}^+(W_d)$. By Theorem [GR17, 1.8], the map

$$\alpha : \mathcal{M}^{\theta_d}(W_d; \hat{l}_{W_d}^+) \longrightarrow \Omega^\infty \text{MT}\theta_d \quad (12)$$

is an isomorphism on the first integral homology group, if regarded onto the path component hit, and given that $g(W_d) \geq 5$.

We will compute the first homology group of this path component by computing that of the basepoint component $H_1(\Omega_0^\infty \text{MT}\theta_d; \mathbb{Z}) \cong \pi_1(\Omega_0^\infty \text{MT}\theta_d) \cong \pi_1(\text{MT}\theta_d)$, and using that all path components of a loop space are homotopy equivalent; to make these identifications, we are using that the Hurewicz map is an isomorphism – because π_1 of a loop space is abelian –, and that the homotopy groups of a spectrum are the same as those of the base point component of its infinite loop space.

Finally, we will show that G is path connected, so that BG is simply connected.

3.2. Cohomology ring of V_d

We let $x = c_1(\mathcal{O}(1)) \in H^2(\mathbb{CP}^\infty; \mathbb{Z})$, so $H^*(\mathbb{CP}^4; \mathbb{Z}) = \mathbb{Z}[x]/(x^5)$. The inclusion $i: V_d \rightarrow \mathbb{CP}^4$ is 3-connected by the Lefschetz theorem on hyperplane sections (see [AF59], for example), so for $t = i^*(x)$ we have

$$H^0(V_d; \mathbb{Z}) = \mathbb{Z} \quad H^1(V_d; \mathbb{Z}) = 0 \quad H^2(V_d; \mathbb{Z}) = \mathbb{Z}\{t\}.$$

Using the Poincaré duality isomorphisms $-\cap[V_d]: H^k(V_d; \mathbb{Z}) \rightarrow H_{6-k}(V_d; \mathbb{Z})$ we deduce that

$$H^4(V_d; \mathbb{Z}) = \mathbb{Z}\{s\} \quad H^5(V_d; \mathbb{Z}) = 0 \quad H^6(V_d; \mathbb{Z}) = \mathbb{Z}\{u\}$$

where u and s are the Poincaré duals of the preferred generators of $H_0(V_d; \mathbb{Z})$ and $H_2(V_d; \mathbb{Z})$, respectively. So $\langle u, [V_d] \rangle = \langle 1, u \cap [V_d] \rangle = 1 \in H_0(V_d; \mathbb{Z})$.

We claim that $t^3 = d \cdot u$, that is

$$\langle t^3, [V_d] \rangle = \langle x^3, i_*[V_d] \rangle = d. \quad (13)$$

For this, notice that under Poincaré duality $D: H^k(\mathbb{CP}^4; \mathbb{Z}) \cong H_{8-k}(\mathbb{CP}^4; \mathbb{Z})$, x^3 corresponds to the fundamental class of $\mathbb{CP}^1 \subset \mathbb{CP}^4$. We may assume that $i: V_d \hookrightarrow \mathbb{CP}^4$ is transverse to $\mathbb{CP}^1$ by the transversality theorem – see [Hir12, Ch. 3]. Then $\mathbb{CP}^1 \cap V_d$ is a manifold of dimension 0, consisting of d many points – so that its fundamental class corresponds to d under $H_0(\mathbb{CP}^4; \mathbb{Z}) \cong \mathbb{Z}$ – and, moreover, $D^{-1}[\mathbb{CP}^1 \cap V_d] = D^{-1}[\mathbb{CP}^1] \cdot D^{-1}i_*[V_d]$. Both things together give 13.

Finally, by the universal coefficients theorem $H^3(V_d; \mathbb{Z})$ is free: the Ext term vanishes because $H_2(V_d; \mathbb{Z}) \cong \mathbb{Z}$. The rank of $H^3(V_d; \mathbb{Z})$ is $4 - \chi(V_d)$, because the Euler characteristic $\chi(V_d)$ equals the alternating sum of the ranks of the cohomology groups. We calculate this Euler characteristic in what follows.

3.3. Chern classes and genus of V_d

We now compute the Chern classes of V_d , which we will use in order to determine the genus and the Moore–Postnikov factorization of the orientation of these manifolds.¹⁴

The normal bundle of the embedding $i: V_d \subset \mathbb{CP}^4$ is $i^*(\mathcal{O}(d))$, because the section of $\mathcal{O}(d)$ defining V_d is transverse to the zero section (cf. the proof of the Pontrjagin–Thom theorem 1.2). Therefore we have a decomposition $TV_d \oplus i^*(\mathcal{O}(d)) = i^*(T\mathbb{CP}^4)$. Taking total Chern classes,

$$c(TV_d)i^*(c(\mathcal{O}(d))) = i^*(c(T\mathbb{CP}^4))$$

¹⁴See Appendix 4.4 for the Definition of the Chern classes.

in the ring $\prod_k H^k(V_d; \mathbb{Z})$ of formal infinite series with the convolution product. Identifying the trivial line bundle $\underline{\mathbb{C}} \cong \text{Hom}_{\mathbb{C}}(\gamma_{1,4}, \gamma_{1,4})$, and the tangent bundle of $\mathbb{CP}^4$,

$$T\mathbb{CP}^4 \oplus \underline{\mathbb{C}} \cong \text{Hom}(\gamma_{1,4}, \underline{\mathbb{C}}^4 / \gamma_{1,4}) \oplus \underline{\mathbb{C}} \cong \mathcal{O}(1)^{\oplus 5}.$$

We reach $c(T\mathbb{CP}^4) = c(T\mathbb{CP}^4 \oplus \underline{\mathbb{C}}) = (1+x)^5$. Using that $c_1(\mathcal{O}(d)) = dx$ we obtain $c(\mathcal{O}(d)) = 1+dx$ (see [MS16, §14]).

With our notation $i^*(x) = t$, the above is

$$c(TV_d) = \frac{(1+t)^5}{1+dt} = \left(\sum_{k=0}^5 \binom{5}{k} t^k \right) (1 - dt + (dt)^2 - (dt)^3 + \dots). \quad (14)$$

For the terms in degree 3 we obtain the formula

$$c_3(TV_d) = \left(\binom{5}{3} - \binom{5}{2}d + \binom{5}{1}d^2 - d^3 \right) t^3.$$

Now, to compute the Euler characteristic $\chi(V_d) = \langle c_3(TV_d), [V_d] \rangle$ we use that $\langle t^3, [V_d] \rangle = d$ by 13, and therefore $\chi(V_d) = (10 - 10d + 5d^2 - d^3)d$.

An upper bound for the genus of V_d is $b_3/2$, where $b_3 = 4 - \chi(V_d)$ is the rank of $H^3(V_d; \mathbb{Z})$. In fact, $g(V_d) = b_3/2$ because by [Wal66, Thm. 1] every closed, smooth, simply connected manifold V of dimension 6 can be written as a connected sum $M \# (S^3 \times S^3)^{\#g}$, for a certain $g \geq 0$, and such that $H_3(M; \mathbb{Z})$ is a finite abelian group. Therefore, $g = b_3/2$ must equal $g(V)$. We conclude that

$$g(V_d) = 2 - (10 - 10d + 5d^2 - d^3)d/2.$$

Remark 3.2. From 14 we see that the first Chern class $c_1(TV_d) = (5-d)t$. From the identity $w_2 \equiv c_1 \pmod{2}$, deduce that $w_2(TV_d)$ is zero if d is odd and non-zero if d is even.

3.4. Moore–Postnikov factorization

We now determine the highly connected structure associated to the orientation $\hat{l}_{W_d}^+$ of W_d , that is the Moore–Postnikov factorization

$$W_d \xrightarrow{l_{W_d}} B_d \xrightarrow{u^+} BSO(6) \quad (15)$$

of $\hat{l}_{W_d}^+$ as a 3-connected cofibration followed by a 3-co-connected fibration; it will depend on the parity of d .

Observe that $W_d = V_d \# (S^3 \times S^3)^{\#g}$ is simply connected by the Seifert–van Kampen theorem, and that $\pi_2(W_d) \cong \mathbb{Z}$ using the Hurewicz theorem and that the quotient map $W_d \rightarrow V_d \vee (S^3 \times S^3)^{\#g}$ is an homology isomorphism in degrees $0 < * < 6$. On the other hand, $BSO(6)$ is simply connected by definition, and $BSO(6) \rightarrow BSO$ is 6-connected so that $\pi_2(BSO(6)) \cong \mathbb{Z}/2$ and $\pi_3(BSO(6)) \cong 0$, by the Bott periodicity theorem. Therefore, such maps l_{W_d} and u^+ must take the following form on

homotopy groups.

$$\begin{aligned}
0 = \pi_1(W_d) &\xrightarrow{\cong} \pi_1(B_d) \\
\mathbb{Z} = \pi_2(W_d) &\xrightarrow{\cong} \pi_2(B_d) \\
\pi_3(B_d) &\xrightarrow{\cong} \pi_3(BSO(6)) = 0 \\
\pi_i(B_d) &\xrightarrow{\cong} \pi_i(BSO(6))
\end{aligned}$$

for all $i > 3$. Such a description of u^+ suggests to us to consider the following candidates for it:

$$B\text{Spin}(6) \times K(\mathbb{Z}, 2) \xrightarrow{pr} B\text{Spin}(6) \longrightarrow BSO(6) \quad (16)$$

$$B\text{Spin}^c(6) \longrightarrow BSO(6). \quad (17)$$

For example, the reason to give the first candidate is that by definition, $\text{Spin}(6) \rightarrow \text{SO}(6)$ is the connected double cover. Moreover, since the only index 2 subgroup of $\pi_1(\text{SO}(6)) \cong \mathbb{Z}/2$ is the zero subgroup, it must be that $\pi_1(\text{Spin}(6)) = 0$ so it is in fact the universal cover. Therefore $B\text{Spin}(6) \rightarrow BSO(6)$ is the 2-connected cover; but $\pi_3(BSO(6)) = 0$, so it is also the 3-connected cover. For $B\text{Spin}^c(6)$ see 18.

The difference between these two is that, on π_2 , 16 induces the zero map $\mathbb{Z} \rightarrow \mathbb{Z}/2$, and 17 induces the non-zero map $\mathbb{Z} \rightarrow \mathbb{Z}/2$. To consolidate these candidates, we determine $\pi_2(l_{W_d}^+)$, the map that they must factor together with a map $W_d \rightarrow B_d$ still to be provided.

Under the identification $\pi_2(BSO(6)) \cong \mathbb{Z}/2$, the homomorphism $\pi_2(l_{W_d}^+)$ is the image of the identity under the map

$$\text{Hom}(\pi_2(W_d), \mathbb{Z}/2) \cong H^2(W_d; \mathbb{Z}/2) \longleftarrow H^2(BSO(6); \mathbb{Z}/2) \cong \text{Hom}(\pi_2(BSO(6)), \mathbb{Z}/2) \ni \text{id}$$

given by precomposition with $\pi_2(l_{W_d}^+)$. On the right hand-side, id corresponds to the universal second Stiefel–Whitney class, thus $\pi_2(l_{W_d}^+)$ corresponds to $w_2(TW_d) \in H^2(W_d; \mathbb{Z}/2)$. Therefore, $\pi_2(l_{W_d}^+)$ is zero if and only if $w_2(TW_d)$ is zero.

In turn, $w_2(TW_d)$ is zero if and only if W_d admits a spin structure. Since $S^3 \times S^3$ and S^5 both admit spin structures, we deduce that $W_d = V_d \sharp (S^3 \times S^3)^{\sharp g}$ admits a spin structure if and only if V_d does. By 3.2, we conclude that $w_2(TW_d)$ is zero if d is odd, and it is non-zero if d is even.

Even d case. $w_2(TW_d)$ does not vanish, but is still the reduction modulo 2 of the generator $q^*(t) = c_1(TW_d)$ ¹⁵ of $H^2(W_d; \mathbb{Z}) = [W_d, K(\mathbb{Z}, 2)]$, where $q: W_d \rightarrow V_d \vee (S^3 \times S^3)^{\sharp g}$ is the quotient map. We will also denote $q^*(t)$ by t . Therefore the lower composite below is nullhomotopic, so we may pick a lift to the homotopy pullback

$$\begin{array}{ccccc}
& & B\text{Spin}^c(6) & \longrightarrow & * \\
& \nearrow \text{dashed} & \downarrow & & \downarrow \\
W_d & \xrightarrow{l^+ \times t} & BSO(6) \times K(\mathbb{Z}, 2) & \xrightarrow{w_2 \times \text{red}_2} & K(\mathbb{Z}/2, 2).
\end{array} \quad (18)$$

¹⁵ W_d does indeed admit an almost complex structure, because for 6-dimensional manifolds the only obstruction is the third Stiefel–Whitney class.

This gives a factorization 15 of the form

$$W_d \longrightarrow B\mathrm{Spin}^c(6) \longrightarrow B\mathrm{SO}(6).$$

Indeed, to see that the map $W_d \rightarrow B\mathrm{Spin}^c(6)$ is 3-connected, note that it trivially induces an isomorphism on π_1 and an epimorphism on π_3 . We claim that it induces an isomorphism on π_2 . To prove it, we look at the long exact sequence of the fiber sequence and we see that $B\mathrm{Spin}^c(6) \rightarrow B\mathrm{SO}(6) \times K(\mathbb{Z}, 2)$ induces $\mathrm{red}_2 \times \mathrm{id}: \mathbb{Z} \rightarrow \mathbb{Z}/2 \times \mathbb{Z}$ on π_2 under the identification of these groups. Therefore, the composite with the projection $B\mathrm{Spin}^c(6) \rightarrow K(\mathbb{Z}, 2)$ induces an isomorphism on π_2 . Next, we will show that $\pi_2(t)$ is also an isomorphism, thus the claim will follow.

The map $H_2: [W_d, K(\mathbb{Z}, 2)] \rightarrow \mathrm{Hom}(H_2(W_d; \mathbb{Z}), H_2(K(\mathbb{Z}, 2); \mathbb{Z}))$ is an isomorphism, by the universal coefficient theorem for cohomology. By the Hurewicz theorem, the latter is isomorphic to $\mathrm{Hom}(\pi_2(W_d), \pi_2(K(\mathbb{Z}, 2))) \cong \mathrm{Hom}(\mathbb{Z}, \mathbb{Z})$, and the image of $[t]$ under this composition of isomorphisms between infinite cyclic groups is $\pi_2(t)$; since $[t]$ was chosen to be a generator, $\pi_2(t)$ must also be so. Finally, under the isomorphism $\mathbb{Z} \cong \mathrm{Hom}(\mathbb{Z}, \mathbb{Z})$ that sends an integer to multiplication by it, we see that the possible generators of the latter group are the isomorphisms $\pm \mathrm{id}$.

Odd d case. $w_2(TW_d) = 0$ so TW_d admits a (unique) spin structure $\hat{l}_{W_d}^{\mathrm{Spin}}$, which gives the factorization $W_d \rightarrow B\mathrm{Spin}(6) \rightarrow B\mathrm{SO}(6)$ as the composite of a 2-connected map followed by a 2-co-connected map. As the 3-connected map $W_d \rightarrow B\mathrm{Spin}(6) \times K(\mathbb{Z}, 2)$ we give $\hat{l}_{W_d}^{\mathrm{Spin}} \times t$ which, again, trivially induces an isomorphism on π_1 and an epimorphism on π_3 . We showed in the previous case that $\pi_2(t)$ is an isomorphism. We conclude that that 15 takes the form

$$W_d \longrightarrow B\mathrm{Spin}(6) \times K(\mathbb{Z}, 2) \longrightarrow B\mathrm{SO}(6).$$

3.5. The spectrum $\mathrm{MT}\theta_d$ and its first homotopy group

Even d case. Then $\theta_d: B\mathrm{Spin}^c(6) \rightarrow \mathrm{BO}(6)$ so $\mathrm{MT}\theta_d = \mathrm{MTSpin}^c(6)$.

Odd d case. Then $\theta_d: B\mathrm{Spin}(6) \times K(\mathbb{Z}, 2) \rightarrow \mathrm{BO}(6)$ and $\mathrm{MT}\theta_d = \mathrm{MTSpin}(6) \wedge K(\mathbb{Z}, 2)_+$ the spacewise smash product. Indeed, its $(d+6)$ th space is the Thom space of the d -dimensional bundle pulled back from $\gamma_{6,d}^\perp$ along $\theta_{6,d}: B\mathrm{Spin}(6)_d \times K(\mathbb{Z}, 2) \rightarrow \mathrm{Gr}_6(\mathbb{R}^{6+d})$ – cf. Definition 1.8. This map is the composition of the projection to the first factor and then the map pulled back from $B\mathrm{Spin}(6) \rightarrow \mathrm{BO}(6)$. Therefore the pulled back bundle is a product with the 0-dimensional vector bundle over $K(\mathbb{Z}, 2)$, and the Thom space of the latter is $K(\mathbb{Z}, 2)_+ \wedge S^0 \cong K(\mathbb{Z}, 2)_+$. Finally, the Thom space of a product is the smash product of Thom spaces.

First homotopy groups of $\mathrm{MTSpin}(6) \wedge K(\mathbb{Z}, 2)_+$ Consider the fiber sequence

$$F \longrightarrow \mathrm{MTSpin}(6) \xrightarrow{s} \Sigma^{-6} \mathrm{MSpin}.$$

We are interested in π_1 of the middle term of the fiber sequence below, obtained by smashing the previous with $K(\mathbb{Z}, 2)_+$.

$$F \wedge K(\mathbb{Z}, 2)_+ \longrightarrow \mathrm{MTSpin}(6) \wedge K(\mathbb{Z}, 2)_+ \longrightarrow \Sigma^{-6} \mathrm{MSpin} \wedge K(\mathbb{Z}, 2)_+$$

We identify the groups $\pi_i(\mathrm{MSpin} \wedge K(\mathbb{Z}, 2)_+) = \Omega_i^{\mathrm{Spin}}(K(\mathbb{Z}, 2))$, and we seek understanding of the groups $\pi_i(F \wedge K(\mathbb{Z}, 2)_+) = F_i(K(\mathbb{Z}, 2))$ for $i = 0, 1$. For this, consider the Atiyah–Hirzebruch spectral sequence for F ,

$$E_{p,q}^2 = H_p(K(\mathbb{Z}, 2); \pi_q(F)) \implies F_{p+q}(K(\mathbb{Z}, 2)),$$

which is a strongly convergent half-plane spectral sequence. We have seen in section 2.1 that there is a 3-connected map $\Sigma^{-6} \mathrm{SO} / \mathrm{SO}(6) \rightarrow F$, and consequently F is connective. Therefore the only entries in the lines $p + q = 0$ and $p + q = 1$ are $E_{0,0}^2 \cong \pi_0 F$ and $E_{0,1}^2 \cong \pi_1 F$, respectively. We conclude that these must be summands in $F_*(K(\mathbb{Z}, 2))$ which the retraction of pointed spaces $S^0 \rightarrow K(\mathbb{Z}, 2)_+ \rightarrow S^0$ splits, and they are all the homotopy in these degrees, that is the retraction $F_*(K(\mathbb{Z}, 2)) \rightarrow F_*(\mathrm{pt}) = \pi_* F$ to the split summand $F_*(\mathrm{pt})$ induces an isomorphism in degrees 0 and 1.

We get the following induced maps between the part of interest in the homotopy long exact sequence

$$\begin{array}{ccccccc} \pi_1 F & \longrightarrow & \pi_1(\mathrm{MTSpin}(6) \wedge K(\mathbb{Z}, 2)_+) & \longrightarrow & \Omega_7^{\mathrm{Spin}}(K(\mathbb{Z}, 2)) & \longrightarrow & \pi_0 F \\ \parallel & & \updownarrow & & \updownarrow & & \parallel \\ \pi_1 F & \longrightarrow & \pi_1(\mathrm{MTSpin}(6)) & \xrightarrow{\pi_1(s)} & \Omega_7^{\mathrm{Spin}} & \longrightarrow & \pi_0 F. \end{array}$$

By Lemma 2.5, $\pi_1(s)$ is surjective, and by Lemma 2.8, $\pi_1(s)$ is injective. Therefore, the two outer maps in the bottom exact row are zero. We learn that the same is true for the two outer maps in the top row, and therefore that

$$\pi_1(\mathrm{MTSpin}(6) \wedge K(\mathbb{Z}, 2)_+) \xrightarrow{\cong} \Omega_7^{\mathrm{Spin}}(K(\mathbb{Z}, 2)).$$

The Atiyah–Bott–Shapiro map $\mathrm{MSpin} \rightarrow ko$ is 8-connected, so it induces an isomorphism of homology groups $\Omega_7^{\mathrm{Spin}}(K(\mathbb{Z}, 2)) \cong ko_7(K(\mathbb{Z}, 2))$.

We show that this group is zero arguing with the Atiyah–Hirzebruch spectral sequence for ko

$$E_{p,q}^2 = H_p(K(\mathbb{Z}, 2); \pi_q ko) \implies ko_{p+q}(K(\mathbb{Z}, 2)).$$

Since $K(\mathbb{Z}, 2) \simeq \mathbb{C}P^\infty$ admits a CW structure with a single cell in each even degree, its homology is free of rank one in each even degree. We deduce that the spectral sequence is concentrated in the even columns, and its entries are known by the Bott periodicity theorem ($\pi_i(ko) = \pi_i(\Omega^\infty ko) = \pi_i(BO \times \mathbb{Z})$). In the line $p + q = 7$ of the E^2 page we have

$$\begin{aligned} E_{0,7}^2 &\cong \pi_7 ko = 0 \\ E_{2,5}^2 &\cong \pi_5 ko = 0 \\ E_{4,3}^2 &\cong \pi_3 ko = 0 \\ E_{6,1}^2 &\cong \pi_1 ko = \mathbb{Z}/2. \end{aligned}$$

We show that the differential $d^2: H_8(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z} \rightarrow H_6(\mathbb{C}P^\infty; \mathbb{Z}/2) \cong \mathbb{Z}/2$ hitting $E_{6,1}^2$ is non-zero and therefore $ko_7(K(\mathbb{Z}, 2)) = 0$. Such differentials from the 0th row are induced by the map $k_0: H\mathbb{Z} \rightarrow \Sigma^2 H(\mathbb{Z}/2)$ – the k_0 -invariant of ko . See 4.2 –, which gives an element in $H^2(H\mathbb{Z}; \mathbb{Z}/2)$.

Serre showed [Ser53, §10] that the reduction mod 2 homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/2$ induces a surjective linear map $H^*(H\mathbb{Z}/2; \mathbb{Z}/2) \rightarrow H^*(H\mathbb{Z}; \mathbb{Z}/2)$ from the Steenrod algebra, which in degree 2 identifies both sides with a one dimensional $\mathbb{Z}/2$ -vector space generated by the Steenrod square Sq^2 . Therefore, k_0 is either zero or given by the composition $H\mathbb{Z} \rightarrow H(\mathbb{Z}/2) \rightarrow \Sigma^2 H(\mathbb{Z}/2)$ of red_2 , the reduction mod 2, and Sq^2 .

The map k_0 is non-zero, and therefore we identify it with $Sq^2 \circ \text{red}_2$. Indeed, since the map $S \rightarrow \text{MSpin}$ induced by $EO \rightarrow BO\langle 3 \rangle = BO\langle 2 \rangle$ is 3-connected, and the Atiyah–Hirzebruch spectral sequence is natural, for the range in question we may consider the spectral sequence for S instead of ko , and applying it to $H(\mathbb{Z}/2)$ it takes the form

$$E_{p,q}^2 = H_p(H(\mathbb{Z}/2); \pi_q S) \implies \pi_{p+q}(H(\mathbb{Z}/2)).$$

Then k_0 induces the differential $d^2: E_{2,0}^2 \rightarrow E_{0,1}^2 \cong \mathbb{Z}/2$ – for this isomorphism, $\pi_1(S) \cong \mathbb{Z}/2$ and then $E_{0,1}^2$ is the degree zero part of the $\mathbb{Z}/2$ -Steenrod algebra. But $\pi_1(H(\mathbb{Z}/2)) = 0$ so $E_{0,1}^2$ must vanish and therefore d^2 be non-zero, because all other differentials involving $E_{0,1}^2$ exit the first quadrant, where the spectral sequence is supported. ([Tei93, p. 751])

To prove that the map $d^2 = \pi_8(k_0 \wedge K(\mathbb{Z}, 2)_+)$ is non-zero it suffices to show that the homomorphism $Sq^2: H^6(\mathbb{C}P^\infty; \mathbb{Z}/2) \rightarrow H^8(\mathbb{C}P^\infty; \mathbb{Z}/2)$ is non-zero, which we do by taking $n = 3$ and $i = 1$ in the formula that follows. Let x , of degree 2, be a generator $H^*(\mathbb{C}P^\infty; \mathbb{Z}/2) \cong \mathbb{Z}/2[x]$. The total Steenrod square is $Sq(x) = Sq^1(x) + Sq^2(x) = x + x^2$, because of the axioms $Sq^1 = 1$, Sq^2 is squaring, and Sq^n vanish on elements of degree smaller than n . The Cartan formula gives $Sq(x^n) = Sq(x)^n = x^n(1+x)^n = \sum_i \binom{n}{i} x^{n+i}$, and from this we conclude that $Sq^{2i}(x^n) = \binom{n}{i} x^{n+i}$.

First homotopy group of $\text{MTSpin}^c(6)$. The map

$$\pi_1(\text{MTSpin}^c(6)) \xrightarrow{\cong} \pi_7(\text{MSpin}^c)$$

is surjective by Lemma 2.7, and injective by Lemma 2.9. By the Pontrjagin–Thom theorem 1.2, $\pi_7(\text{MTSpin}^c) \cong \Omega_7^{\text{Spin}^c}$. This cobordism group has been calculated [BG87, p. 5] and it is zero. We conclude that

$$\pi_1(\text{MTSpin}^c(6)) = 0.$$

3.6. Path components of homotopy automorphisms

Recall that $G \leq \text{hAut}(u^+)$ is the submonoid of those path components that preserve $\hat{l}_{W_d}$ up to orientation preserving diffeomorphism of W_d . We show that G is path connected. For this, let $f \in G$ be arbitrary, and we show that it is connected to id by a path in G , that is $\text{id} \sqcup f: S^0 \rightarrow G$ extends along $S^0 \rightarrow D^1$.

This lifting problem is equivalent to the following, by adjunction.

$$\begin{array}{ccc} S^0 \times B_d & \xrightarrow{\text{id} \sqcup f} & B_d \\ \downarrow & \nearrow \text{dashed} & \downarrow u^+ \\ D^1 \times B_d & \xrightarrow{pr} B_d & \xrightarrow{u^+} BSO(6). \end{array}$$

The part of the diagram below the dashed arrow imposes the condition that the self maps of B_d commute with u^+ ; any such lift is automatically a weak homotopy equivalence, as it is a homotopy of such.

We will solve a different lifting problem which will imply the above. The arbitrary element f of G preserves the tangential θ_d -structure $\hat{l}_{W_d}$ up to orientation preserving diffeomorphism of W_d . That is, there exists an orientation preserving diffeomorphism ϕ of W_d such that $\hat{f} \circ \hat{l}_{W_d} \circ d\phi$ is in the same path component of the space of bundle maps as $\hat{l}_{W_d}$. For the underlying maps of base spaces this means that $f \circ l_{W_d} \circ \phi$ is homotopic to l_{W_d} . We obtain a commutative diagram

$$\begin{array}{ccc} H^2(W_d; \mathbb{Z}) & \xleftarrow[\cong]{l_{W_d}^*} & H^2(B_d; \mathbb{Z}) \\ \downarrow \phi^* & & \uparrow f^* \\ H^2(W_d; \mathbb{Z}) & \xleftarrow[\cong]{l_{W_d}^*} & H^2(B_d; \mathbb{Z}) \end{array}$$

We claim that ϕ^* maps the generator t to itself, and therefore f^* must also be the identity homomorphism. Indeed, analyzing the cohomology $H^*(W_d; \mathbb{Z})$ and its cup product structure via the quotient map $W_d \rightarrow V_d \vee (S^3 \times S^3)^{\sharp g}$, we see that $t^3 = d \cdot u'$ where u' is a generator for $H^6(W_d; \mathbb{Z}) \cong H_6(W_d; \mathbb{Z})$, in fact the dual of the fundamental class and therefore ϕ^* must send it to itself. Suppose for contradiction that ϕ^* sends t to $-t$. But then $\phi^*(t^3) = \phi^*(t)^3 = -d \cdot u'$.

We have shown that $t \circ f$ is homotopic to t . Therefore, we can form the following diagram which commutes strictly in the $BSO(6)$ factor, and up to homotopy in the $K(\mathbb{Z}, 2)$ factor.

$$\begin{array}{ccc} S^0 \times B_d & \xrightarrow{\text{id} \sqcup f} & B_d \\ \downarrow & & \downarrow u^+ \times t \\ D^1 \times B_d & \xrightarrow{pr} B_d \xrightarrow{u^+ \times t} & BSO(6) \times K(\mathbb{Z}, 2). \end{array} \quad (19)$$

We will show the existence of a lift which makes the lower triangle commute strictly, and the upper triangle commute up to homotopy over $BSO(6)$, that is up to a homotopy that commutes strictly with the maps to $BSO(6)$. This will conclude the proof.

The fiber of the map $u^+ \times t$ is a $K(\mathbb{Z}/2, 1)$ in both the cases that d is even or odd. Indeed, for the odd case, the fiber of the product of fibrations $u^+ \times t: B\text{Spin}(6) \times K(\mathbb{Z}, 2) \rightarrow BSO(6) \times K(\mathbb{Z}, 2)$ is the product of the fibers $K(\mathbb{Z}/2, 1) \times \{pt\}$. The even case is [18](#).

3.6.1 Obstruction theory

Suppose we have two spaces $A \hookrightarrow W$, a fibration $B \rightarrow Z$, and maps between them forming a commutative square as below. We ask when can we find a lift such as the dashed arrow. Suppose that the fibration fits in a fiber sequence $B \rightarrow Z \rightarrow K$, so B is the pullback along the fibration $ev_0: P_b K \rightarrow K$ sending a path ending at the basepoint b of K to the starting point of the path.

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \longrightarrow & P_b K \\ \downarrow & \nearrow & \downarrow p & & \downarrow ev_0 \\ W & \xrightarrow{g} & Z & \xrightarrow{k} & K. \end{array}$$

A lift amounts to a nullhomotopy of $W \rightarrow K$ extending that of $A \rightarrow K$, equivalently a map $CW \rightarrow K$ extending the given $W \cup CA \rightarrow K$. For example, if we write $CA = (A \times [0, 1]) / \sim$ by identifying all points in $A \times \{1\}$, then the map $A \rightarrow B$ is recovered from $c_A: CA \rightarrow K$ as $a \mapsto (p(f(a)), \gamma_a)$ where γ_a is the path $c_A(a, -)$. If $W \cup CA \rightarrow K$ is nullhomotopic, we claim that we can obtain such an extension using the homotopy extension property of the pair $(CW, W \cup CA)$. Note that if $K = K(G, n+1)$ is an Eilenberg–Mac Lane space, we can determine if the map is nullhomotopic by determining if the class it represents $\omega_n \in H^{n+1}(W \cup CA; G) \cong H^{n+1}(W, A; G)$ is zero or not.

Now suppose that the left square only commutes up to a homotopy $H: A \times [0, 1] \rightarrow Z$. We explain how the same procedure gives a lift which commutes up to homotopy. We use the homotopy H to consider a map $W \cup_h CA = W \cup A \times [0, 1] \cup CA \rightarrow K$ from the homotopy pushout. Suppose that this map is nullhomotopic, and as before we obtain an extension $c_W: CW \rightarrow K$, thus a map $W \rightarrow B$, $w \mapsto (g(w), \gamma'_w)$. It is clear that $W \rightarrow B \rightarrow Z$ equals g . Now, we give a homotopy from $A \rightarrow W \rightarrow B$ to f . Notice that if $w \in A \hookrightarrow W$, then $\gamma'_w = H(a, -) * \gamma_a$ is the concatenation, up to reparametrization. The homotopy $A \times [0, 1] \rightarrow B$ sends (a, s) to $(H(a, s), \tilde{H}(a, s) * \gamma_a)$, where $\tilde{H}(a, s)$ is the path that travels along $H(a, -)$ at double speed during times in between 0 and $1-s$, and is constant $H(a, 1)$ for the rest of the time.

In our situation 19, $K = K(\mathbb{Z}/2, 2)$ and $\omega_1 \in H^2(D^1 \times B_d, S^0 \times B_d; \mathbb{Z}/2)$. This cohomology group is zero. Indeed, we look at the long exact sequence of the pair in cohomology and we see that $S^0 \times B_d \rightarrow D^1 \times B_d$ induces a monomorphism – precisely, the diagonal map into the product –, therefore $H^1(S^0 \times B_d; \mathbb{Z}/2)$ surjects onto the group in question. But $S^0 \times B_d$ is simply connected, so this first cohomology group is zero.

As explained above we obtain a lift which makes the bottom triangle commute and the upper triangle commute up to homotopy. Moreover, in this case we may pick the homotopy H so that $u^+ \circ H: S^0 \times B_d \times [0, 1] \rightarrow B_d \rightarrow BSO(6)$ does not vary in t , because of the strict commutativity in the factor $BSO(6)$. Consequently, the homotopy we gave above witnessing homotopy commutativity of the upper triangle commutes strictly with the maps to $BSO(6)$.

We conclude that G is path connected. All we have used about the manifold $W = W_d$ is the assumptions of Theorem 1.10 (including 6-dimensional) and that an orientation preserving diffeomorphism of W acts as the identity on its second integral cohomology group, so that $t \circ f$ is homotopic to $t: B \rightarrow K(\pi_2(W), 2)$.

3.7. Summing up

Consider the fiber sequence

$$\mathcal{M}^{\theta_d}(W; \hat{l}_{W_d}) \longrightarrow \mathcal{M}^+(W_d, \hat{l}_{W_d}^+) \longrightarrow BG. \quad (20)$$

For such a fiber sequence $F \rightarrow E \rightarrow B$, the Serre spectral sequence gives exact sequences

$$\begin{aligned} 0 \longrightarrow E_{0,1}^\infty \longrightarrow H_1(E; \mathbb{Z}) \longrightarrow E_{1,0}^\infty = E_{1,0}^2 \longrightarrow 0 \\ E_{2,0}^2 \xrightarrow{d^2} E_{0,1}^2 \longrightarrow E_{0,1}^3 = E_{0,1}^\infty \longrightarrow 0. \end{aligned}$$

Putting these together we obtain an exact sequence $E_{0,1}^2 \rightarrow H_1(E; \mathbb{Z}) \rightarrow E_{1,0}^\infty = E_{1,0}^2$, that is

$$H_0(B; H_1(F; \mathbb{Z})) \longrightarrow H_1(E; \mathbb{Z}) \longrightarrow H_1(B; H_0(F; \mathbb{Z})).$$

In our case 20, the base space is simply connected by the discussion above, and the fiber is path connected. Therefore the exact sequences take the form

$$H_1(\mathcal{M}^{\theta_d}(W_d; \hat{l}_{W_d}); \mathbb{Z}) \longrightarrow H_1(\mathcal{M}^+(W_d; \hat{l}_{W_d}^+); \mathbb{Z}) \longrightarrow 0$$

The group at the left identifies with $\pi_1(\text{MT}\theta_d)$ by Theorem [GR17, 1.8]. In sections 3.5 and 3.5 we showed that $\pi_1(\text{MT}\theta_d) = 0$ if d is odd and if d is even, respectively. Recall the homotopy equivalence 1.7 $\mathcal{M}^+(W_d; \hat{l}_{W_d}^+) \simeq \text{BDiff}^+(W_d)$. We conclude that

$$H_1(\text{BDiff}^+(W_d); \mathbb{Z}) = 0.$$

This finishes the proof of Proposition 3.1.

4. APPENDIX

Here is material that has been referenced throughout the thesis. We have put it apart for a more fluent development of the main text. The topics are topological monoids 4.1, the Thom isomorphism 4.3, the Atiyah–Hirzebruch spectral sequence, and characteristic classes 4.4.

4.1. Homotopy orbits of topological monoid actions.

Definition 4.1. A *topological monoid* is a triple (M, μ, η) consisting of a topological space M , and continuous maps $\mu: M \times M \rightarrow M$ and $\eta: * \rightarrow M$ that satisfy associativity and unitality, that is, they fit in commutative diagrams

$$\begin{array}{ccc} M \times M \times M & \xrightarrow{\text{id} \times \mu} & M \times M \\ \downarrow \mu \times \text{id} & & \downarrow \mu \\ M \times M & \xrightarrow{\mu} & M \end{array} \qquad \begin{array}{ccccc} * \times M & \xrightarrow{\eta \times \text{id}} & M \times M & \xleftarrow{\text{id} \times \eta} & M \times * \\ & \searrow & \downarrow \mu & \swarrow & \\ & & M & & \end{array}$$

We say that M is *group-like* if $(\pi_0(M), \pi_0(\mu), \pi_0(\eta))$ is a group.

Let M be a topological monoid, let X a space with a left action $\alpha: M \times X \rightarrow X$ (a continuous map satisfying associativity $\alpha \circ (\text{id} \times \alpha) = \alpha \circ (\mu \times \text{id})$ and unitality $\alpha \circ (\eta \times \text{id}) = \text{id}$), and let Y be a space with a right action $\beta: Y \times M \rightarrow Y$ (defined similarly). We may form a simplicial space

$$B_\bullet(Y, M, X): [p] \longmapsto Y \times M^p \times X = B_p(Y, M, X)$$

with face maps $d_i: B_p(Y, M, X) \rightarrow B_{p-1}(Y, M, X)$ induced by μ for $0 < i < p$, and by β and α for $i = 0, p$, respectively. Given a simplicial space $B_\bullet$, we can form a topological space called the *fat geometric realization* by gluing standard p -simplices $\Delta^p \subset \mathbb{R}^{p+1}$, one for each p -simplex of $B_\bullet$, according to the face maps of $B_\bullet$. We refer to [ER19] for precise definitions and details.

Definition 4.2. Let M be a topological monoid, and X a space with a left action $M \times X \rightarrow X$. The space of *homotopy orbits* $X // M$ is defined to be the fat geometric realization $||B_\bullet(*, M, X)||$.

Definition 4.3. A commutative square of spaces

$$\begin{array}{ccc} X_1 & \xrightarrow{k_1} & Y_1 \\ f \downarrow & & \downarrow g \\ X_0 & \xrightarrow{k_0} & Y_0 \end{array}$$

is called a *homotopy pullback* if for each basepoint $x \in X_0$, the map $\text{hofib}_x(f) \rightarrow \text{hofib}_{k_0(x)}(g)$ induced by k_0 and k_1 is a weak homotopy equivalence. If X_0 is contractible, we call it a (*homotopy*) *fiber sequence*.

Given a diagram $X \xrightarrow{g} A \xleftarrow{f} Y$, we may factor $f : Y \xrightarrow{\sim} P \rightarrow A$ as a weak equivalence followed by a fibration, by taking $P = \{(y, \lambda) \in Y \times \text{Map}([0, 1], A) \mid \lambda(1) = f(y)\}$ and the evaluation at 0 map to A , for example. Then, we refer to the strict pullback $X \times_A P$ as the *homotopy pullback* of $X \rightarrow A \leftarrow Y$. It is indeed a homotopy pullback square as in Definition 4.3. If $X = *$ is a point, we refer to it as the *homotopy fiber* of f over $*$.

The following holds, for example, if M is a group-like topological monoid.

Theorem 4.1. Let M be a topological monoid, and X be a space with a left M -action $\alpha : M \times X \rightarrow X$. Suppose that for every $m \in M$ its action $\alpha_m : X \rightarrow X$ is a weak equivalence. Then there is a fiber sequence

$$X \longrightarrow X // M \longrightarrow * // M = BM.$$

Proof. Consider the map of simplicial spaces $\pi : X // M \rightarrow * // M$. We show that for each $p \geq 1$ and each $0 \leq i \leq p$, the square formed with the face maps

$$\begin{array}{ccc} M^p \times X & \xrightarrow{d_i} & M^{p-1} \times X \\ \pi_p \downarrow & & \downarrow \pi_{p-1} \\ M^p & \xrightarrow{d_i} & M^{p-1} \end{array}$$

is a homotopy pullback. Then the result follows by [ER19, 2.12].

The relevant case is $i = p$. Then the face map d_p of $X // M$ is $\text{id}^{(p-1)} \times \alpha$, and for $* // M$ it is the projection pr to the first $p - 1$ factors. Let $m^{(p)} = (m_1, \dots, m_p) \in M^p$ be arbitrary. The map induced between the homotopy fibers is

$$\begin{aligned} \text{hofib}_{m^{(p)}} &\simeq \{(n^{(p)}, x, \gamma) \in M^p \times X \times \text{Map}([0, 1], M^p) \mid \gamma(0) = n^{(p)}, \gamma(1) = m^{(p)}\} \\ &\quad \downarrow \text{id}^{(p-1)} \times \alpha \times pr_* \\ \text{hofib}_{d_i(m^{(p)})} &\simeq \{(n^{(p-1)}, x, \gamma) \in M^{p-1} \times X \times \text{Map}([0, 1], M^{p-1}) \mid \gamma(0) = n^{(p-1)}, \gamma(1) = d_i(m^{(p)})\}. \end{aligned}$$

Every point $(n^{(p)}, x, \gamma)$ in $\text{hofib}_{m^{(p)}}$ comes with a path γ from $n^{(p)}$ to $m^{(p)}$ in M^p , and similarly for $\text{hofib}_{d_i(m^{(p)})}$. Contracting both spaces along these paths, onto $\{m^{(p)}\} \times X \times *$ and $\{d_i(m^{(p)})\} \times X \times *$, respectively, we see that the map between homotopy fibers is homotopic to the action $\alpha_{m_p} : X \rightarrow X$ of m_p , which by assumption is a weak homotopy equivalence. $\square$

Write $BM = ||B_\bullet(*, G, *)||$, $EM = ||B_\bullet(-, M, M)||$ and observe that there is a canonical map $M \rightarrow \Omega BM$. The map $EM \rightarrow *$ is a weak equivalence by [ER19, 1.12]. Therefore, in the situation of Theorem 4.1, it follows from the long exact sequence in homotopy that $M \rightarrow \Omega BM$ is a weak equivalence, so

$$\pi_{i+1}(BM) \cong \pi_i(M). \quad (21)$$

In particular, suppose that $M = G$ is a discrete group-like topological monoid, i.e., a group. Then BG is an Eilenberg–Mac Lane $K(G, 1)$ space, that is

$$\pi_i(BG) \cong \begin{cases} G & i = 1 \\ 0 & i \neq 1 \end{cases}$$

If G is abelian, then BG is again an abelian group and this construction can be iterated. Using 21, we see that $B^n(G)$ is an Eilenberg–Mac Lane space $K(G, n)$.

Path components of orbit spaces. Let G be a group-like topological monoid with an action $G \times X \rightarrow X$ on a space X , equivalently, α is a map of topological monoids $G \rightarrow \mathbf{hAut}(X)$ to the space of weak equivalences of X . Since the action is continuous, it induces an action of $\pi_0(G)$ on $\pi_0(X)$. The map $\pi_0(X) \rightarrow \pi_0(X // G)$ induces a bijection $\pi_0(X)/\pi_0(G) \sim \pi_0(X // G)$ – with the model given above as the two-sided bar construction, this is clear because there is a 1-simplex of $X // G$ between two 0-simplices $x_0, x_1 \in X$ precisely if there exists $m \in M$ such that $m \cdot x_0 = x_1$, and the 1-simplex is then $(m, x_0) \in M \times X$.

The path component of the G -orbit of a point x of X . Let $X_{x \cdot \pi_0(G)}$ be the union of those path components of X that are represented in $\pi_0(X)$ by some point of the $\pi_0(G)$ -orbit of x . Then the action of $\pi_0(G)$ restricts to $X_{x \cdot \pi_0(G)}$ and establishes weak equivalence between every pair of its path components. Note that $X_{x \cdot \pi_0(G)} // G$ is path connected, and the inclusion $X_x \subset X_{x \cdot \pi_0(G)}$ induces a weak equivalence $X_x // G_x \sim X_{x \cdot \pi_0(G)} // G$, where X_x is the path component of X which contains x , and $G_x \subset G$ is the pullback

$$\begin{array}{ccc} G_x & \longrightarrow & G \\ \downarrow & & \downarrow -x \\ [x] & \longrightarrow & \pi_0(X) \end{array}$$

in the category of sets. That is, $G_x \subset G$ consists of those components of G that stabilize $[x] \in \pi_0(X)$. We see that it is a submonoid.

4.2. Atiyah–Hirzebruch spectral sequence

We refer to [Wei95] for background on spectral sequences. Let E be a spectrum, and X be a space or a spectrum. Consider the Postnikov tower of E , together with the successive cofibers, that is the k -invariants of E .

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \tau_{\leq q+1} E & \longrightarrow & \tau_{\leq q} E & \longrightarrow & \tau_{\leq q-1} E \longrightarrow \cdots \\ & & & & \downarrow k_q & & \downarrow k_{q-1} \\ & & \cdots & & \Sigma^{q+2} \pi_{q+1} E & & \Sigma^{q+1} \pi_q E \longrightarrow \cdots \end{array}$$

Smashing with X we obtain another tower and form the exact couple of bigraded abelian groups

$$\begin{array}{ccc}
 \bigoplus_{p,q} \pi_{p+q}(X \wedge \tau_{\leq q} E) & \xrightarrow{i} & \bigoplus_{p,q} \pi_{p+q}(X \wedge \tau_{\leq q-1} E) \\
 & \nwarrow k \quad \nearrow j & \\
 & \bigoplus_{p,q} \pi_{p+q}(X \wedge \Sigma^{q+1} H \pi_q E) &
 \end{array}$$

where p and q vary over the integers and k decreases the degree in π_* by 1. We identify

$$E_{p,q}^2 = \pi_{p+q}(X \wedge \Sigma^{q+1} H \pi_q E) \cong H_{p-1}(X; \pi_q E).$$

We reindex the spectral sequence so that $E_{p,q}^2 \cong H_p(X \wedge; \pi_q E)$, and then the d^2 differential is given by the desuspension of $j \circ k$, that is

$$\Sigma^q H \pi_q E \longrightarrow \tau_{\leq q} E \xrightarrow{j=k_q} \Sigma^{q+2} H \pi_{q+1} E.$$

Example 4.2. Let $E = ko$. Then, by the Bott periodicity theorem, its Postnikov tower starts as

$$\begin{array}{ccccccccc}
 \cdots & \longrightarrow & \tau_{\leq 4} ko & \longrightarrow & \tau_{\leq 3} ko & \longrightarrow & \tau_{\leq 2} ko & \longrightarrow & \tau_{\leq 1} ko & \longrightarrow & \tau_{\leq 0} ko & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow k_3 & & \downarrow & & \downarrow k_1 & & \downarrow k_0 & & \downarrow \\
 & & 0 & & \Sigma^5 H\mathbb{Z} & & 0 & & \Sigma^3 H\mathbb{Z}/2 & & \Sigma^2 H\mathbb{Z}/2 & & \Sigma H\mathbb{Z}
 \end{array}$$

and continues according to the homotopy groups of ko given by Bott periodicity. The d^2 differentials from the 0th row in the spectral sequence are induced by $d^2 = k_0: H\mathbb{Z} = \tau_{\leq 0} ko \rightarrow \Sigma^2 H\mathbb{Z}/2$.

4.3. The Thom isomorphism

Let R be a ring. We prove the Thom isomorphism for HR -orientable stable vector bundles. We first prove it for unstable vector bundles and afterwards show how to reduce the stable case to the unstable one. In this section, all (co)homology is with coefficients in R , so $H(-) = H(-; R)$.

Let $\xi: V \rightarrow X$ be a real vector bundle of rank $d \geq 0$, and consider the fiber bundle $p: S^\xi \rightarrow X$ obtained by applying one-point compactification to each fiber of ξ . Denote by $i_x: S_x^d \rightarrow S^\xi$ the inclusion of the fiber over $x \in X$; we may use the same notation for the composite with the quotient map $q: S^\xi \rightarrow \text{Th}(\xi)$. Taking the product $p \times q$ and precomposing with the diagonal gives a map $S^\xi \rightarrow X \times \text{Th}(\xi)$ which induces a cup product on cohomology, and also descends to a map

$$\text{Th}(\xi) \rightarrow X_+ \wedge \text{Th}(\xi).$$

The latter induces the cup and cap products

$$\begin{aligned}
 H^*(X) \otimes \tilde{H}^*(\text{Th}(\xi)) &\longrightarrow \tilde{H}^*(\text{Th}(\xi)), & x \otimes y &\mapsto p^*(x) \cup y \\
 \tilde{H}_*(\text{Th}(\xi)) \otimes \tilde{H}^*(\text{Th}(\xi)) &\longrightarrow H_*(X), & z \otimes y &\mapsto p_*(z \cap y).
 \end{aligned}$$

(We are omitting the quotient map in the notation.)

Recall that an R -orientation, or Thom class, of ξ is a cohomology class $u \in \tilde{H}^d(\text{Th}(\xi))$ such that, for every point $x \in X$, $i_x^*(u)$ is a generator for the free R -module $H^n(S_x^d)$.

Remark 4.3. Fixing u , these products are Hom-dual maps.

Theorem 4.4 (Thom isomorphism). *Let $u \in \tilde{H}^d(\text{Th}(\xi))$ be an R -orientation for a rank d real vector bundle $\xi: V \rightarrow X$. Let*

$$\phi: H^k(X) \longrightarrow \tilde{H}^{k+d}(\text{Th}(\xi))$$

be given by $\phi(y) = p^(y) \cup u$. Then ϕ is an isomorphism of R -modules.*

Proof. Let $b \in X$. The R -orientation provides an isomorphism between the local coefficient systems $x \mapsto H^d(S_x^d)$ and the constant $H^d(S_b^d)$. The Serre spectral sequence for $S^d = S_b^d \rightarrow S^\xi \rightarrow X$ takes the form

$$H^p(S^\xi; H^q(S^d)) \implies H^{p+q}(S^\xi).$$

The entries in the spectral sequence remain unchanged until the $(d+1)$ th page, the differentials of which fit in the Gysin sequence

$$\dots \longrightarrow H^{i-(d+1)}(X; H^d S^d) \xrightarrow{d_{d+1}} H^i(X; H^0 S^d) \xrightarrow{p^*} H^i(S^\xi) \xrightarrow{p_!} H^{i-d}(X; H^d S^d) \xrightarrow{d_{d+1}} \dots \quad (22)$$

More precisely,

$$\begin{aligned} p_! : H^i(S^\xi) &\longrightarrow E_\infty^{i-d,d} \hookrightarrow E_2^{i-d,d} = H^{i-d}(X; H^d S^d) \\ p^* : H^i(X; H^0 S^d) &= E_2^{i,0} \longrightarrow E_\infty^{i,0} \hookrightarrow H^i(S^\xi). \end{aligned}$$

We proceed to further describe $p_!$. Consider the image of the R -orientation u under the composite $p_! \circ q^*: \tilde{H}^d(\text{Th}(\xi)) \rightarrow H^d(S^\xi) \rightarrow H^0(X; H^d S^d)$, which we will denote by $p_!(u)$. The spectral sequence admits a multiplicative structure induced by a cup product in the second page, which in turn induces one in the following pages because the differentials are derivations with respect to this product. We also have the cup product structure on $H^*(S^\xi)$, and all these agree in the sense that they are involved in the following commutative diagram.

$$\begin{array}{ccccc} E_2^{i-d,0} & \times & E_2^{0,d} & \xrightarrow{\cup} & E_2^{i-d,d} \\ \downarrow & & \uparrow & & \uparrow \\ E_\infty^{i-d,0} & \times & E_\infty^{0,d} & \xrightarrow{\cup} & E_\infty^{i-d,d} \\ \downarrow & & \uparrow & & \uparrow \\ H^{i-d}(S^\xi) & \times & H^d(S^\xi) & \xrightarrow{\cup} & H^i(S^\xi) \end{array}$$

Notice that the downward vertical composite is p^* and the upward vertical composites are $p_!$. Commutativity says that if $y \in E_2^{i-d,0}$, then $p_!(p^*(y) \cup u) = y \cup p_!(u)$. This fully describes the image of $p_!$ because multiplication by $p_!(u)$ is an isomorphism $E_2^{i-d,0} \rightarrow E_2^{i-d,d}$. Indeed, by assumption $p_!(u) \in E_2^{0,d} = H^0(X; H^d S^d)$ is a generator and the product is induced by the cup product structure on $H^*(S^d)$.

Finally, notice that the section $s: X \rightarrow S^\xi$ of p^* hitting the points at infinity splits the exact sequence 22 and therefore identifies the quotient $H^{i-d}(X; H^d S^d)$ with the corresponding one in the split long exact sequence induced by the cofiber sequence $q \circ s: X \rightarrow S^\xi \rightarrow \text{Th}(\xi)$, that is the composite $p_! \circ q^*: \tilde{H}^i(\text{Th}(\xi)) \rightarrow H^{i-d}(X; H^d S^d)$ is an isomorphism. We may further compose this map with the isomorphism $\cup p_!(u)^{-1}: H^{i-d}(X; H^d S^d) \rightarrow H^{i-d}(X; H^0 S^d)$. As before, we omit q^* in the notation and we can describe this composite as $p^*(y) \cup u \mapsto y \cup p_!(u) \mapsto y$, therefore its inverse ϕ is given by $\phi(y) = p^*(y) \cup u$. $\square$

Let $V : X \rightarrow \mathbb{Z} \times BO$, and suppose that X is path connected; otherwise we consider one path component at a time. Let $d \in \mathbb{Z}$ be its rank. Consider the map $\cdot u : H^*(X) \rightarrow H^*(\mathbf{Th}(V))$ in [1](#). We show how it takes the form of an unstable Thom isomorphism as above by proving in what follows that the HR-cohomology of $\mathbf{Th}(V)$ is that of a Thom space far enough in the spectrum.

Homotopy and (co)homology of Thom spectra. We show that the system giving the colimit $\pi_k(\mathbf{Th}(V)) = \operatorname{colim}_n \pi_{n+k}(\mathbf{Th}(V)_n)$ is eventually constant, and conclude by the Hurewicz theorem. A map in this system is the composition

$$\pi_{n+k}(\mathbf{Th}(V)_n) \rightarrow \pi_{n+1+k}(\Sigma \mathbf{Th}(V)_n) \rightarrow \pi_{n+1+k}(\mathbf{Th}(V)_{n+1})$$

of the suspension homomorphism and the structure map σ_n of the spectrum. We prove that the connectivity of these maps is of the form $n + f(n)$ where $f : \mathbb{N} \rightarrow \mathbb{N}$ is some function tending to infinity as n does. It then follows that for any fixed k , these maps are isomorphisms for all big enough n .

The n th space of the spectrum $\mathbf{Th}(V)$ is $\mathbf{Th}(v_{n+d}^* \gamma_{n+d})$, where $v_{n+d} : X_{n+d} \rightarrow BO(n+d)$ is the pullback along $BO(n+d) \rightarrow BO$. This space is $(n+d)$ -connected because the bundle is of rank $n+d$ – see [MS16, 18.1]. The Freudenthal suspension theorem deals with the connectivity of the suspension homomorphism.

Regarding the connectivity of the map σ_n , the map $BO(n+d) \rightarrow BO(n+1+d)$ is $(n+d)$ -connected, as it can be modeled by a fibration with fiber S^{n+d} . Then the map $X_{n+d} \rightarrow X_{n+1+d}$ is also $(n+d)$ -connected, because it is pulled-back from the fibration $BO(n+d) \rightarrow BO(n+1+d)$, so the pullback agrees with the homotopy pullback and their homotopy fibers are weakly equivalent. Then the map of Thom spaces

$$\sigma_n : \Sigma \mathbf{Th}(f_{n+d}^* \gamma_{n+d} \oplus \epsilon) \longrightarrow \mathbf{Th}(f_{n+1+d}^* \gamma_{n+1+d})$$

is $((n+d) + (n+1+d))$ -connected, that is the connectivity of the map of base spaces plus the dimension of the bundles. This finishes the proof.

Remark 4.5. As a consequence, the cohomology groups of a Thom spectrum are well-defined as in [23](#) below, because the system stabilizes by the argument above and the Hurewicz theorem.

We conclude that

$$H^k(X) = \lim_n H^k(X_{n+d}) \xrightarrow{\cong} H^{k+d}(\mathbf{Th}(V)) = \lim_n H^{n+k+d}(\mathbf{Th}(f_{n+d}^* \gamma_{n+d})) \quad (23)$$

by the Thom isomorphism [4.4](#).

4.4. Characteristic classes

We define the Stiefel–Whitney classes of a real vector bundle, the Euler class of an orientable vector bundle, and the Chern classes of complex vector bundles. We then present an axiomatic characterization of these.

Stiefel Whitney classes. Let $\xi: V \rightarrow X$ be a real d -dimensional vector bundle, and let $P \rightarrow X$ be the associated principal $O(d)$ -bundle. Let $k \leq d$. We are going to define the k th Stiefel–Whitney class of V . Consider the fiber bundle $p_k: P/O(k-1) \rightarrow X$ with fiber the coset space $O(d)/O(k-1)$. (If $k = d$, then this bundle is isomorphic to the sphere bundle of the universal vector bundle γ_d over $BO(d)$.) We are going to use the Serre spectral sequence, so let us study the homotopy of this fiber. We have a fibration sequence $O(k) \rightarrow O(k+1) \rightarrow O(k+1)/O(k) \cong S^k$, thus the first map is $(k-1)$ -connected and it follows that the middle vertical map below is $(k-1)$ -connected

$$\begin{array}{ccccc} O(k-1) & \longrightarrow & O(k) & \longrightarrow & O(k)/O(k-1) \\ \parallel & & \downarrow & & \downarrow \\ O(k-1) & \longrightarrow & O(d) & \longrightarrow & O(d)/O(k-1). \end{array}$$

Since the homotopy fibers of the middle and right vertical maps are weakly equivalent – because the fiber of the map between vertical fibers is weakly equivalent to the fiber of the map between horizontal fibers $\text{id}: O(k-1) \rightarrow O(k-1)$ – we conclude that the right vertical map

$$S^{k-1} \longrightarrow O(d)/O(k-1)$$

is $(k-1)$ -connected.

Therefore, the cohomological Serre spectral sequence with $\mathbb{Z}/2$ -coefficients for the fiber bundle $P/O(k-1) \rightarrow X$ satisfies $E_2^{p,q} = 0$ for all $0 < q < k-1$, and

$$E_2^{0,k-1} \cong H^0(X; \mathbb{Z}/2) \otimes_{\mathbb{Z}/2} H^{k-1}(O(d)/O(k-1); \mathbb{Z}/2)$$

using the Universal Coefficient Theorem. By Lemma 2.4 and the Hurewicz theorem, the right factor $H^{k-1}(O(d)/O(k-1); \mathbb{Z}/2)$ is of rank 1; let ι be the generator. The first non-zero differential involving this entry is $d_k: E_2^{0,k-1} \rightarrow E_2^{k,0} = H^k(X; \mathbb{Z}/2)$; it is called a *transgression*.

Definition 4.4. The k th Stiefel–Whitney class $w_k(\xi)$ of the vector bundle $\xi: V \rightarrow X$ is the cohomology class $d_k(1 \otimes \iota) \in H^k(X; \mathbb{Z}/2)$.

Euler class. We can follow the same procedure as above with coefficients in $\mathbb{Z}$ when ξ is orientable, and taking $d = k$ gives the definition of the Euler class $e(\xi) \in H^d(X; \mathbb{Z})$ of the spherical fiber bundle $S^{d-1} \rightarrow S(V) \rightarrow X$, or we simply say the Euler class of ξ . Observe that e reduces modulo 2 to the Stiefel–Whitney class $w_d(\xi)$.

Chern Classes. Let $\xi: V \rightarrow X$ be a complex d -dimensional vector bundle. By considering $U(d)$ instead of $O(d)$ and arguing similarly, we can define the Chern classes $c_k(\xi) \in H^{2k}(X; \mathbb{Z})$. Note that in this case there is a choice of generator for

$$H^{2k-1}(U(d)/U(k-1); \mathbb{Z}) \xrightarrow{\cong} H^{2k-1}(U(k)/U(k-1); \mathbb{Z}) \cong \mathbb{Z}$$

to be made. Complex vector bundles admit a canonical orientation, which provides a generator for this group; we choose the opposite of this generator. Note that the Chern class $c_k(\xi)$ reduces modulo 2 to $w_{2k}(\xi)$.

The formal sum

$$c(\xi) = 1 + c_1(\xi) + c_2(\xi) + \cdots + c_d(\xi)$$

in the ring $\prod_k H^k(X; \mathbb{Z})$ of formal infinite series with the convolution product is called the *total Chern class* of ξ . It is a unit in this ring because the leading term $c_0(\xi) = 1$ is a unit in $H^0(X; \mathbb{Z})$. Axiom 3 in Theorem 4.7 can be expressed as

$$c(\xi \oplus \eta) = c(\xi)c(\eta) \in \prod_k H^k(X; \mathbb{Z}).$$

These characteristic classes have an axiomatic description given in the following two theorems, the proofs of which are given in [MS16]. We write $w_k = w_k(\gamma_d)$ and $c_k = c_k(\gamma_d^{\mathbb{C}})$ for a universal vector bundle γ_d , and a universal complex vector bundle $\gamma_d^{\mathbb{C}}$, respectively.

Theorem 4.6. *The Stiefel–Whitney classes satisfy and are uniquely characterized by the following four axioms.*

1. To each (real) vector bundle $\xi : V \rightarrow X$ there corresponds a sequence of cohomology classes $w_k(\xi) \in H^k(X; \mathbb{Z}/2)$, $k \in \mathbb{N}$, such that $w_0(\xi) = 1 \in H^0(X; \mathbb{Z}/2)$, and $w_k(\xi) = 0$ if k is greater than the dimension of ξ .
2. Naturality. If $f : X \rightarrow Y$ is covered by a bundle map $\xi \rightarrow \eta$, then $w_k(\xi) = f^*(w_k(\eta))$.
3. Product theorem. If ξ and η are vector bundles over the same base space, then

$$w_k(\xi \oplus \eta) = \sum_{i=0}^k w_i(\xi) \cup w_{k-i}(\eta).$$

4. For the tautological line bundle γ_1^1 over $\mathbb{R}P^1$, $w_1(\gamma_{1,1}) \neq 0$.

Moreover, $H^*(\text{Gr}_d(\mathbb{R}^\infty); \mathbb{Z}/2)$ is the polynomial algebra $\mathbb{Z}/2[w_1, \dots, w_d]$.

Theorem 4.7. *The Chern classes satisfy and are uniquely characterized by the following four axioms.*

1. To each complex vector bundle $\xi : V \rightarrow X$ there corresponds a sequence of cohomology classes $c_k(\xi) \in H^{2k}(X; \mathbb{Z})$, $k \in \mathbb{N}$, such that $c_0(\xi) = 1 \in H^0(X; \mathbb{Z})$, and $c_k(\xi) = 0$ if k is greater than the dimension of ξ .
2. Naturality. If $f : X \rightarrow Y$ is covered by a bundle map $\xi \rightarrow \eta$, then $c_k(\xi) = f^*(c_k(\eta))$.
3. Product theorem. If ξ and η are vector bundles over the same base space, then

$$c_k(\xi \oplus \eta) = \sum_{i=0}^k c_i(\xi) \cup c_{k-i}(\eta).$$

4. For the complex line bundle $\mathcal{O}(1) = \text{Hom}_{\mathbb{C}}(\gamma_{1,1}^{\mathbb{C}}, \underline{\mathbb{C}})$ over $\mathbb{C}P^1$, $c_1(\mathcal{O}(1))$ is the canonical generator for $H^2(\mathbb{C}P^1; \mathbb{Z})$ given by the orientation.

Moreover, $H^*(\text{Gr}_d(\mathbb{C}^\infty); \mathbb{Z})$ is the polynomial algebra $\mathbb{Z}[c_1, \dots, c_d]$.

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